

Research Methods in Psychology

Research Methods in Psychology

4th edition

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Contents

Acknowledgements	ix
About this Book	xi
About the Authors of the Current Edition	xvi
Preface	xviii
Chapter I. <u>The Science of Psychology</u>	
1. Methods of Knowing	3
2. Understanding Science	6
3. Goals of Science	10
4. Science and Common Sense	12
5. Experimental and Clinical Psychologists	15
6. Key Takeaways and Exercises	19
Chapter II. <u>Overview of the Scientific Method</u>	
7. A Model of Scientific Research in Psychology	25
8. Finding a Research Topic	28
9. Generating Good Research Questions	36
10. Developing a Hypothesis	40
11. Designing a Research Study	45
12. Analyzing the Data	49
13. Drawing Conclusions and Reporting the Results	52
14. Key Takeaways and Exercise	54
Chapter III. <u>Research Ethics</u>	
15. Moral Foundations of Ethical Research	59
16. From Moral Principles to Ethics Codes	65
17. Putting Ethics Into Practice	74
18. Key Takeaways and Exercises	79

Chapter IV. Psychological Measurement

19. Understanding Psychological Measurement	83
20. Reliability and Validity of Measurement	92
21. Practical Strategies for Psychological Measurement	99
22. Key Takeaways and Exercises	105

Chapter V. Experimental Research

23. Experiment Basics	109
24. Experimental Design	117
25. Experimentation and Validity	125
26. Practical Considerations	130
27. Key Takeaways and Exercises	138

Chapter VI. Non-Experimental Research

28. Overview of Non-Experimental Research	143
29. Correlational Research	148
30. Complex Correlation	157
31. Qualitative Research	163
32. Observational Research	169
33. Key Takeaways and Exercises	179

Chapter VII. Survey Research

34. Overview of Survey Research	185
35. Constructing Surveys	188
36. Conducting Surveys	198
37. Key Takeaways and Exercises	204

Chapter VIII. Quasi-Experimental Research

38. One-Group Designs	209
39. Non-Equivalent Groups Designs	215
40. Key Takeaways and Exercises	219

Chapter IX. Factorial Designs

41. Setting Up a Factorial Experiment	223
42. Interpreting the Results of a Factorial Experiment	229
43. Key Takeaways and Exercises	238

Chapter X. Single-Subject Research

44. Overview of Single-Subject Research	241
45. Single-Subject Research Designs	244
46. The Single-Subject Versus Group “Debate”	254
47. Key Takeaways and Exercises	259

Chapter XI. Presenting Your Research

48. American Psychological Association (APA) Style	263
49. Writing a Research Report in American Psychological Association (APA) Style	272
50. Other Presentation Formats	287
51. Key Takeaways and Exercises	293

Chapter XII. Descriptive Statistics

52. Describing Single Variables	297
53. Describing Statistical Relationships	309
54. Expressing Your Results	321
55. Conducting Your Analyses	332
56. Key Takeaways and Exercises	337

Chapter XIII. Inferential Statistics

57. Understanding Null Hypothesis Testing	343
58. Some Basic Null Hypothesis Tests	350
59. Additional Considerations	366
60. From the “Replicability Crisis” to Open Science Practices	374
61. Key Takeaways and Exercises	382

Glossary	385
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References	417
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Rajiv, Carrie, and Dana (May 2019)



Rajiv S. Jhangiani, Carrie Cuttler, & Dana C. Leighton

About this Book

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The first Canadian edition (published in 2013) was authored by Rajiv S. Jhangiani (Kwantlen Polytechnic University) and licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 3.0 License](https://creativecommons.org/licenses/by-nc-sa/3.0/). Revisions included the addition of a table of contents, changes to Chapter 3 (Research Ethics) to include a contemporary example of an ethical breach and to reflect Canadian ethical guidelines and privacy laws, additional information regarding online data collection in Chapter 9 (Survey Research), corrections of errors in the text and formulae, spelling changes from US to Canadian conventions, the addition of a cover page, and other necessary formatting adjustments.

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of anonymity, elaborating on the Belmont Report (the principles of respect for persons and beneficence were added), and adding a link to a clip dispelling the myth that vaccines cause autism. Sections from Chapter 4 (on defining theories and hypotheses) were moved to Chapter 2 and the remainder of the previous Chapter 4 (on phenomenon, theories, and hypotheses) was cut. Chapter 5 was reorganized by moving the sections on four types of validity, manipulation checks, and placebo effects to later in the chapter. Descriptions of single factor two-level designs, single factor multi-level designs, matched-groups designs, order effects, and random counterbalancing were added to Chapter 5 and the concept of statistical validity was expanded upon. Chapter 6 was also reorganized by moving sections describing correlation coefficients from Chapters 2 and 12 to Chapter 6. The section of the book on complex correlation was also moved to Chapter 6 and the section on quasi-experiments was moved from Chapter 6 to its own chapter (Chapter 8). The categories of non-experimental research described in Chapter 6 were changed to cross-sectional, correlational, and observational research. Chapter 6 was further expanded to describe cross-sectional studies, partial correlation, simple regression, the use of regression to make predictions, case studies, participant observation, disguised and undisguised observation, and structured observation. The terms independent variable and dependent variable as used in the context of regression were changed to predictor variable and outcome/criterion variable respectively. A distinction between proportionate stratified sampling and disproportionate stratified sampling was added to Chapter 7. The section on quasi-experimental designs was moved to its own chapter (Chapter 8) and was elaborated upon to include instrumentation and testing as threats to internal validity of one-group pretest-posttest designs, and to include sections describing the one-group posttest only design, pretest-posttest nonequivalent groups design, interrupted time-series with nonequivalent groups design, pretest-posttest design with switching replication, and switching replication with treatment removal designs. The section of Chapter 9 on factorial designs was split into two sections and the remainder of the chapter was moved or cut. Further, examples of everyday interactions were added and a description of simple effects was added to Chapter 9. The section on case studies that appeared in Chapter 10 was edited and moved to Chapter 6. Further, labels were added to multiple-baseline across behaviours, settings, and participants designs, and a concluding paragraph on converging evidence was added to Chapter 10. Only minor edits were made to the remaining chapters (Chapters 11, 12, and 13).

This fourth edition (published in 2019) was co-authored by Rajiv S. Jhangiani (Kwantlen Polytechnic University), Carrie Cuttler (Washington State University), and Dana C. Leighton (Texas A&M University–Texarkana) and is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](#). Revisions throughout the current edition include changing the chapter and section numbering system to better accommodate adaptations that remove or reorder chapters; continued reversion from the Canadian edition; general grammatical edits; replacement of “he/she” to “they” and “his/her” to “their”; removal or update of dead links; embedded videos that were not embedded; moved key takeaways and exercises from the end of each chapter section to the end of each chapter; a new cover design. In addition, the following revisions were made to specific chapters:

- Chapter 1:
 - Updated list of empirically supported therapies.
- Chapter 2:
 - Added description of follow-up research by Drews, Pasupathi, and Strayer (2004) demonstrating

that cell phone conversations while driving carry a greater risk than conversations with a passenger

- Added the term meta-analysis along with a definition of this term
- Replaced terms men and women with males and females
- Updated the description of the number of records returned with different search terms to a broader description of the relative number of records (that will not change as more articles are added to PsychINFO)
- Replaced the term “operationally define” variables with a more general statement about measuring variables since the term operational definition is not formally defined until later in the text
- Added a citation for Zajonc’s (1965) research
- Added a brief description of factors (i.e., small sample size, stringent alpha level) that increase the likelihood of a Type II error.
- Chapter 3:
 - Removed titles of tables in references to tables
 - Added statement that many people, including children, have died as a result of people avoiding the MMR vaccine
 - Added a statement about self-plagiarizing being unethical and provided an example of submitting the same assignment in multiple classes
 - Explained the respect for persons principle
 - Revised the levels of IRB review to match terminology used in federal regulations
 - Footnotes for references were made actual footnotes in Pressbooks
- Chapter 4:
 - Removed potentially offensive or stigmatizing examples
 - Clarified definition of levels of measurement
 - Added citations for the various scales described
 - Added further description of why IQ is measured on an interval scale
 - Added descriptions of the indicators of central tendency that are appropriate to compute and report for each of the scales of measure (nominal, ordinal, interval, ratio)
 - Added a paragraph on operationally defining the construct that reviews the process of transferring a conceptual definition to something that can be directly observed and measured
 - Added brief description of PsycTESTS and link to these tests
 - Removed the statement that family and friends can serve as good pilot subjects
- Chapter 5:
 - Clarified the distinction between independent and dependent variables
 - Moved up the discussion of a control condition
 - Briefly discussed research ethics within the description of the study by Guéguen & de Gail (2003)
 - More clearly defined a power analysis and emphasized the importance of conducting one
 - Referenced confounds within the discussion of internal validity
 - Noted that within-subjects experiments require fewer participants
 - Removed duplicate reference
 - Added citations
 - Updated language

- Chapter 6:
 - Clarified when non-experimental approaches are appropriate
 - Added information about Milgram's non-experimental study of obedience to authority
 - Added a discussion of cross-sectional, longitudinal, and cross-sequential studies
 - Revised organization of non-experimental approaches
 - Removed description of experimenter-selected independent variable
 - Specified types of variables that may be measured in correlational research
 - Added an example of a correlational study that uses categorical variables
 - Added a factor analysis table
 - Listed more examples of nonstatistical data analysis techniques
 - Added a table to summarize some differences between quantitative and qualitative research
 - Described some group dynamics and personality characteristics that might influence participation in focus groups
 - Discussed Festinger's research on cognitive dissonance that used disguised participant observation
 - Described the Hawthorne effect
 - Added an example of a study that used structured observation within a laboratory environment
- Chapter 7:
 - Clarified language concerning data collection methods vs. research designs
 - Mentioned randomizing the order of presentation of questions as another way of reducing response order effects
 - Explained reverse coding
 - Described additional types of non-probability sampling
 - Reiterated the importance of conducting a power analysis
 - Added common online data collection sites
- Chapter 8:
 - Discussed how the inclusion of a control group rules out threats to internal validity within a one-group design study
- Chapter 9:
 - Clarified discussion of non-experimental factorial designs.
- Chapter 10: No substantive changes
- Chapter 11:
 - Added regional psychology association conferences to list of conferences
 - Condensed and clarified discussion of final manuscripts
 - Updated discussion of open sharing of results to acknowledge some journals that require open data
 - Added explanation of person-first language
- Chapter 12:
 - Corrected erroneous APA style recommendations and added references to specific Publication Manual sections
 - Standardized the use of the terms "figure" and "chart" to better correspond with APA style

- Minor changes to discussion of poster formatting
- Moved list of conferences to end of discussion to not break up the material
- Chapter 13:
 - Defined p-hacking and clarified discussion of p-hacking
 - Made definition of p -value more technically correct

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Preface

Psychology, like most other sciences, has its own set of tools to investigate the important research questions of its field. Unlike other sciences that are older and more mature, psychology is a relatively new field and, like an adolescent, is learning and changing rapidly. Psychology researchers are learning and changing along with the emerging science. This textbook introduces students to the fundamental principles of what it is like to think like a psychology researcher in the contemporary world of psychology research.

Historically, psychology developed practices and methods based on the established physical sciences. Unlike physical sciences, psychology had to grapple with the inherent variation among its subjects: people. To better account for this, we developed some practices and statistical methods that we (naïvely) considered to be foolproof. Over time we established a foundation of research findings that we considered solid.

In recent years, psychology's conversation has shifted to an introspective one, looking inward and re-examining the knowledge that we considered foundational. We began to find that some of that unshakable foundation was not as strong as we thought; some of the bedrock findings in psychology were being questioned and failed to be upheld in fuller scrutiny. As many introspective conversations do, this one caused a crisis of faith.

Psychologists are now questioning if we really know what we thought we knew or if we simply got lucky. We are struggling to understand how what we choose to publish and not publish, what we choose to report and not report, and how we train our students as researchers is having an effect on what we call "knowledge" in psychology. We are beginning to question whether that knowledge represents real behaviour and mental processes in human beings, or simply represents the effects of our choice of methods. This has started a firestorm among psychology researchers, but it is one that needs to play out. For a book aimed at novice psychology undergraduates, it is tempting to gloss over these issues and proclaim that our "knowledge" is "truth." That would be a disservice to our students though, who need to be critical questioners of research. Instead of shying away from this controversy, this textbook invites the reader to step right into the middle of it.

With every step of the way, the research process in psychology is fraught with decisions, trade-offs, and uncertainty. We decide to study one variable and not another; we balance the costs of research against its benefits; we are uncertain whether our results will replicate. Every step is a decision that takes us in a different direction and closer to or further from the truth. Research is not an easy route to traverse, but we hope this textbook will be a hiking map that can at least inspire the direction students can take, and provide some absolute routes to begin traveling.

As we wrote at the beginning of this preface, psychology is a young science. Like any adolescent, psychology is grappling with its identity as a science, learning to use better tools, understanding the importance of transparency, and is having more open conversations to improve its understanding of human behaviour. We will grow up and mature together. It is an exciting time to be part of that growth as psychology becomes a more mature science.

CHAPTER I

THE SCIENCE OF PSYCHOLOGY

Many people believe that women tend to talk more than men—with some even suggesting that this difference has a biological basis. One widely cited estimate is that women speak 20,000 words per day on average and men speak only 7,000. This claim seems plausible, but is it *true*? A group of psychologists led by Matthias Mehl decided to find out. They checked to see if anyone had actually tried to count the daily number of words spoken by women and men. No one had. So these researchers conducted a study in which female and male college students (369 in all) wore audio recorders while they went about their lives. The result? The women spoke an average of 16,215 words per day and the men spoke an average of 15,669—an extremely small difference that could easily be explained by chance. In an article in the journal *Science*, these researchers summed up their findings as follows: “We therefore conclude, on the basis of available empirical evidence, that the widespread and highly publicized stereotype about female talkativeness is unfounded” (Mehl, Vazire, Ramirez-Esparza, Slatcher, & Pennebaker, 2007, p. 82)¹.

Psychology is usually defined as the scientific study of human behavior and mental processes, and this example illustrates the features that make it scientific. In this chapter, we look closely at these features, review the goals of psychology, and address several basic questions that students often have about it. Who conducts scientific research in psychology? Why? Does scientific psychology tell us anything that common sense does not? Why should I bother to learn the scientific approach—especially if I want to be a clinical psychologist and not a researcher? These are extremely good questions, and answering them now will provide a solid foundation for learning the rest of the material in your course.

Notes

1. Mehl, M. R., Vazire, S., Ramirez-Esparza, N., Slatcher, R. B., & Pennebaker, J. W. (2007). Are women really more talkative than men? *Science*, 317, 82.

I. Methods of Knowing

Learning Objectives

1. Describe the 5 methods of acquiring knowledge
2. Understand the benefits and problems with each.

Take a minute to ponder some of what you know and how you acquired that knowledge. Perhaps you know that you should make your bed in the morning because your mother or father told you this is what you should do, perhaps you know that swans are white because all of the swans you have seen are white, or perhaps you know that your friend is lying to you because she is acting strange and won't look you in the eye. But should we trust knowledge from these sources? The methods of acquiring knowledge can be broken down into five categories each with its own strengths and weaknesses.

Intuition

The first method of knowing is intuition. When we use our intuition, we are relying on our guts, our emotions, and/or our instincts to guide us. Rather than examining facts or using rational thought, intuition involves believing what feels true. The problem with relying on intuition is that our intuitions can be wrong because they are driven by cognitive and motivational biases rather than logical reasoning or scientific evidence. While the strange behavior of your friend may lead you to think s/he is lying to you it may just be that s/he is holding in a bit of gas or is preoccupied with some other issue that is irrelevant to you. However, weighing alternatives and thinking of all the different possibilities can be paralyzing for some people and sometimes decisions based on intuition are actually superior to those based on analysis (people interested in this idea should read Malcolm Gladwell's book *Blink*)¹.

Authority

Perhaps one of the most common methods of acquiring knowledge is through authority. This method involves accepting new ideas because some authority figure states that they are true. These authorities include parents, the media, doctors, Priests and other religious authorities, the government, and professors. While in an ideal world we should be able to trust authority figures, history has taught us otherwise and many instances of atrocities against humanity are a consequence of people unquestioningly following

authority (e.g., Salem Witch Trials, Nazi War Crimes). On a more benign level, while your parents may have told you that you should make your bed in the morning, making your bed provides the warm damp environment in which mites thrive. Keeping the sheets open provides a less hospitable environment for mites. These examples illustrate that the problem with using authority to obtain knowledge is that they may be wrong, they may just be using their intuition to arrive at their conclusions, and they may have their own reasons to mislead you. Nevertheless, much of the information we acquire is through authority because we don't have time to question and independently research every piece of knowledge we learn through authority. But we can learn to evaluate the credentials of authority figures, to evaluate the methods they used to arrive at their conclusions, and evaluate whether they have any reasons to mislead us.

Rationalism

Rationalism involves using logic and reasoning to acquire new knowledge. Using this method premises are stated and logical rules are followed to arrive at sound conclusions. For instance, if I am given the premise that all swans are white and the premise that this is a swan then I can come to the rational conclusion that this swan is white without actually seeing the swan. The problem with this method is that if the premises are wrong or there is an error in logic then the conclusion will not be valid. For instance, the premise that all swans are white is incorrect; there are black swans in Australia. Also, unless formally trained in the rules of logic it is easy to make an error. Nevertheless, if the premises are correct and logical rules are followed appropriately then this is sound means of acquiring knowledge.

Empiricism

Empiricism involves acquiring knowledge through observation and experience. Once again many of you may have believed that all swans are white because you have only ever seen white swans. For centuries people believed the world is flat because it appears to be flat. These examples and the many visual illusions that trick our senses illustrate the problems with relying on empiricism alone to derive knowledge. We are limited in what we can experience and observe and our senses can deceive us. Moreover, our prior experiences can alter the way we perceive events. Nevertheless, empiricism is at the heart of the scientific method. Science relies on observations. But not just any observations, science relies on structured observations which is known as systematic empiricism.

The Scientific Method

The **scientific method** is a process of systematically collecting and evaluating evidence to test ideas and answer questions. While scientists may use intuition, authority, rationalism, and empiricism to generate new ideas they don't stop there. Scientists go a step further by using systematic empiricism to make careful

observations under various controlled conditions in order to test their ideas and they use rationalism to arrive at valid conclusions. While the scientific method is the most likely of all of the methods to produce valid knowledge, like all methods of acquiring knowledge it also has its drawbacks. One major problem is that it is not always feasible to use the scientific method; this method can require considerable time and resources. Another problem with the scientific method is that it cannot be used to answer all questions. As described in the following section, the scientific method can only be used to address empirical questions. This book and your research methods course are designed to provide you with an in-depth examination of how psychologists use the scientific method to advance our understanding of human behavior and the mind.

Notes

1. Gladwell, M. E. (2005). *Blink: The power of thinking without thinking*. (9th ed.). New York: Little, Brown & Co.

2. Understanding Science

Learning Objectives

1. Define science.
2. Describe the three fundamental features of science.
3. Explain why psychology is a science.
4. Define pseudoscience and give some examples.

What Is Science?

Some people are surprised to learn that psychology is a **science**. They generally agree that astronomy, biology, and chemistry are sciences but wonder what psychology has in common with these other fields. Before answering this question, however, it is worth reflecting on what astronomy, biology, and chemistry have in common with *each other*. It is clearly not their subject matter. Astronomers study celestial bodies, biologists study living organisms, and chemists study matter and its properties. It is also not the equipment and techniques that they use. Few biologists would know what to do with a radio telescope, for example, and few chemists would know how to track a moose population in the wild. For these and other reasons, philosophers and scientists who have thought deeply about this question have concluded that what the sciences have in common is a general approach to understanding the natural world. Psychology is a science because it takes this same general approach to understanding one aspect of the natural world: human behavior.

Features of Science

The general scientific approach has three fundamental features (Stanovich, 2010)¹. The first is **systematic empiricism**. Empiricism refers to learning based on observation, and scientists learn about the natural world systematically, by carefully planning, making, recording, and analyzing observations of it. As we will see, logical reasoning and even creativity play important roles in science too, but scientists are unique in their insistence on checking their ideas about the way the world is against their systematic observations. Notice, for example, that Mehl and his colleagues did not trust other people's stereotypes or even their own informal observations. Instead, they systematically recorded, counted, and compared the number of words spoken by a large sample of women and men. Furthermore, when their systematic observations turned out to conflict with people's stereotypes, they trusted their systematic observations.

The second feature of the scientific approach—which follows in a straightforward way from the first—is that it is concerned with **empirical questions**. These are questions about the way the world actually is and, therefore, can be answered by systematically observing it. The question of whether women talk more than men is empirical in this way. Either women really do talk more than men or they do not, and this can be determined by systematically observing how much women and men actually talk. Having said this, there are many interesting and important questions that are not empirically testable and that science is not in a position to answer. Among these are questions about values—whether things are good or bad, just or unjust, or beautiful or ugly, and how the world *ought* to be. So although the question of whether a stereotype is accurate or inaccurate is an empirically testable one that science can answer, the question—or, rather, the value judgment—of whether it is wrong for people to hold inaccurate stereotypes is not. Similarly, the question of whether criminal behavior has a genetic basis is an empirical question, but the question of what actions ought to be considered illegal is not. It is especially important for researchers in psychology to be mindful of this distinction.

The third feature of science is that it creates **public knowledge**. After asking their empirical questions, making their systematic observations, and drawing their conclusions, scientists publish their work. This usually means writing an article for publication in a professional journal, in which they put their research question in the context of previous research, describe in detail the methods they used to answer their question, and clearly present their results and conclusions. Increasingly, scientists are opting to publish their work in open access journals, in which the articles are freely available to all – scientists and nonscientists alike. This important choice allows publicly-funded research to create knowledge that is truly public.

Publication is an essential feature of science for two reasons. One is that science is a social process—a large-scale collaboration among many researchers distributed across both time and space. Our current scientific knowledge of most topics is based on many different studies conducted by many different researchers who have shared their work publicly over many years. The second is that publication allows science to be self-correcting. Individual scientists understand that, despite their best efforts, their methods can be flawed and their conclusions incorrect. Publication allows others in the scientific community to detect and correct these errors so that, over time, scientific knowledge increasingly reflects the way the world actually is.

A good example of the self-correcting nature of science is the “[Many Labs Replication Project](#)” – a large and coordinated effort by prominent psychological scientists around the world to attempt to replicate findings from 13 classic and contemporary studies (Klein et al., 2013)². One of the findings selected by these researchers for replication was the fascinating effect, first reported by Simone Schnall and her colleagues at the University of Plymouth, that washing one’s hands leads people to view moral transgressions—ranging from keeping money inside a found wallet to using a kitten for sexual arousal—as less wrong (Schnall, Benton, & Harvey, 2008)³. If reliable, this effect might help explain why so many religious traditions associate physical cleanliness with moral purity. However, despite using the same materials and nearly identical procedures with a much larger sample, the “Many Labs” researchers were unable to replicate the original finding (Johnson, Cheung, & Donnellan, 2013)⁴, suggesting that the original finding may have stemmed from the relatively small sample size (which can lead to unreliable results) used in the original study. To be clear, at this stage we are still unable to definitively conclude that the handwashing effect does not exist; however,

the effort that has gone into testing its reliability certainly demonstrates the collaborative and cautious nature of scientific progress.

For more on the replication crisis in psychology see: <http://nobaproject.com/modules/the-replication-crisis-in-psychology>

Science Versus Pseudoscience

Pseudoscience refers to activities and beliefs that are claimed to be scientific by their proponents—and may appear to be scientific at first glance—but are not. Consider the theory of biorhythms (not to be confused with sleep cycles or circadian rhythms that do have a scientific basis). The idea is that people's physical, intellectual, and emotional abilities run in cycles that begin when they are born and continue until they die. Allegedly, the physical cycle has a period of 23 days, the intellectual cycle a period of 33 days, and the emotional cycle a period of 28 days. So, for example, if you had the option of when to schedule an exam, you would want to schedule it for a time when your intellectual cycle will be at a high point. The theory of biorhythms has been around for more than 100 years, and you can find numerous popular books and websites about biorhythms, often containing impressive and scientific-sounding terms like *sinusoidal wave* and *bioelectricity*. The problem with biorhythms, however, is that scientific evidence indicates they do not exist (Hines, 1998)⁵.

A set of beliefs or activities can be said to be pseudoscientific if (a) its adherents claim or imply that it is scientific but (b) it lacks one or more of the three features of science. For instance, it might lack systematic empiricism. Either there is no relevant scientific research or, as in the case of biorhythms, there is relevant scientific research but it is ignored. It might also lack public knowledge. People who promote the beliefs or activities might claim to have conducted scientific research but never publish that research in a way that allows others to evaluate it.

A set of beliefs and activities might also be pseudoscientific because it does not address empirical questions. The philosopher Karl Popper was especially concerned with this idea (Popper, 2002)⁶. He argued more specifically that any scientific claim must be expressed in such a way that there are observations that would—if they were made—count as evidence against the claim. In other words, scientific claims must be **falsifiable**. The claim that women talk more than men is falsifiable because systematic observations could reveal either that they do talk more than men or that they do not. As an example of an unfalsifiable claim, consider that many people who believe in extrasensory perception (ESP) and other psychic powers claim that such powers can disappear when they are observed too closely. This makes it so that no possible observation would count as evidence against ESP. If a careful test of a self-proclaimed psychic showed that she predicted the future at better-than-chance levels, this would be consistent with the claim that she had psychic powers. But if she failed to predict the future at better-than-chance levels, this would also be consistent with the claim because her powers can supposedly disappear when they are observed too closely.

Why should we concern ourselves with pseudoscience? There are at least three reasons. One is that learning about pseudoscience helps bring the fundamental features of science—and their importance—into sharper focus. A second is that biorhythms, psychic powers, astrology, and many other pseudoscientific beliefs

are widely held and are promoted on the Internet, on television, and in books and magazines. Far from being harmless, the promotion of these beliefs often results in great personal toll as, for example, believers in pseudoscience opt for “treatments” such as homeopathy for serious medical conditions instead of empirically-supported treatments. Learning what makes them pseudoscientific can help us to identify and evaluate such beliefs and practices when we encounter them. A third reason is that many pseudosciences purport to explain some aspect of human behavior and mental processes, including biorhythms, astrology, graphology (handwriting analysis), and magnet therapy for pain control. It is important for students of psychology to distinguish their own field clearly from this “pseudo psychology.”

The Skeptic's Dictionary

An excellent source for information on pseudoscience is *The Skeptic's Dictionary* (<http://www.skeptdic.com>). Among the pseudoscientific beliefs and practices you can learn about are the following:

- **Cryptozoology.** The study of “hidden” creatures like Bigfoot, the Loch Ness monster, and the chupacabra.
- **Pseudoscientific psychotherapies.** Past-life regression, rebirthing therapy, and bioscream therapy, among others.
- **Homeopathy.** The treatment of medical conditions using natural substances that have been diluted sometimes to the point of no longer being present.
- **Pyramidology.** Odd theories about the origin and function of the Egyptian pyramids (e.g., that they were built by extraterrestrials) and the idea that pyramids, in general, have healing and other special powers.

Another excellent online resource is *Neurobonkers* (<http://neurobonkers.com>), which regularly posts articles that investigate claims that pertain specifically to psychological science.

Notes

1. Stanovich, K. E. (2010). *How to think straight about psychology* (9th ed.). Boston, MA: Allyn & Bacon.
2. Klein, R. A., Ratliff, K. A., Vianello, M., Adams, R. B., Bahník, S., Bernstein, M. J., . . . Nosek, B. A. (2013). Investigating variation in replicability: A “many labs” replication project. *Social Psychology*, 45(3), 142-152. doi: 10.1027/1864-9335/a000178
3. Schnall, S., Benton, J., & Harvey, S. (2008). With a clean conscience: Cleanliness reduces the severity of moral judgments. *Psychological Science*, 19(12), 1219-1222. doi: 10.1111/j.1467-9280.2008.02227.x
4. Johnson, D. J., Cheung, F., & Donnellan, M. B. (2013). Does cleanliness influence moral judgments? A direct replication of Schnall, Benton, and Harvey (2008). *Social Psychology*, 45(3), 209-215. doi: 10.1027/1864-9335/a000186
5. Hines, T. M. (1998). Comprehensive review of biorhythm theory. *Psychological Reports*, 83, 19-64.
6. Popper, K. R. (2002). *Conjectures and refutations: The growth of scientific knowledge*. New York, NY: Routledge.

3. Goals of Science

Learning Objectives

1. Describe the three goals of science and give an example for each.
2. Distinguish between basic research and applied research.

The Broader Purposes of Scientific Research in Psychology

People have always been curious about the natural world, including themselves and their behavior (in fact, this is probably why you are studying psychology in the first place). Science grew out of this natural curiosity and has become the best way to achieve detailed and accurate knowledge. Keep in mind that most of the phenomena and theories that fill psychology textbooks are the products of scientific research. In a typical introductory psychology textbook, for example, one can learn about specific cortical areas for language and perception, principles of classical and operant conditioning, biases in reasoning and judgment, and people's surprising tendency to obey those in positions of authority. And scientific research continues because what we know right now only scratches the surface of what we *can* know.

The Three Goals of Science

The first and most basic goal of science is **to describe**. This goal is achieved by making careful observations. As an example, perhaps I am interested in better understanding the medical conditions that medical marijuana patients use marijuana to treat. In this case, I could try to access records at several large medical marijuana licensing centers to see which conditions people are getting licensed to use medical marijuana. Or I could survey a large sample of medical marijuana patients and ask them to report which medical conditions they use marijuana to treat or manage. Indeed, research involving surveys of medical marijuana patients has been conducted and has found that the primary symptom medical marijuana patients use marijuana to treat is pain, followed by anxiety and depression (Sexton, Cuttler, Finnell, & Mischley, 2016).¹

The second goal of science is **to predict**. Once we have observed with some regularity that two behaviors or events are systematically related to one another we can use that information to predict whether an event or behavior will occur in a certain situation. Once I know that most medical marijuana patients use marijuana to treat pain I can use that information to predict that an individual who uses medical marijuana likely experiences pain. Of course, my predictions will not be 100% accurate but if the relationship between medical marijuana use and pain is strong then my predictions will have greater than chance accuracy.

The third and ultimate goal of science is **to explain**. This goal involves determining the causes of behavior. For example, researchers might try to understand the mechanisms through which marijuana reduces pain. Does marijuana reduce inflammation which in turn reduces pain? Or does marijuana simply reduce the distress associated with pain rather than reducing pain itself? As you can see these questions tap at the underlying mechanisms and causal relationships.

Basic versus Applied Research

Scientific research is often classified as being either basic or applied. **Basic research** in psychology is conducted primarily for the sake of achieving a more detailed and accurate understanding of human behavior, without necessarily trying to address any particular practical problem. The research of Mehl and his colleagues falls into this category. **Applied research** is conducted primarily to address some practical problem. Research on the effects of cell phone use on driving, for example, was prompted by safety concerns and has led to the enactment of laws to limit this practice. Although the distinction between basic and applied research is convenient, it is not always clear-cut. For example, basic research on sex differences in talkativeness could eventually have an effect on how marriage therapy is practiced, and applied research on the effect of cell phone use on driving could produce new insights into basic processes of perception, attention, and action.

Notes

1. Sexton, M., Cuttler, C., Finnell, J., & Mischley, L (2016). A cross-sectional survey of medical cannabis users: Patterns of use and perceived efficacy. *Cannabis and Cannabinoid Research*, 1, 131-138. doi: 10.1089/can.2016.0007.

4. Science and Common Sense

Learning Objectives

1. Explain the limitations of common sense when it comes to achieving a detailed and accurate understanding of human behavior.
2. Give several examples of common sense or folk psychology that are incorrect.
3. Define skepticism and its role in scientific psychology.

Can We Rely on Common Sense?

Some people wonder whether the scientific approach to psychology is necessary. Can we not reach the same conclusions based on common sense or intuition? Certainly we all have intuitive beliefs about people's behavior, thoughts, and feelings—and these beliefs are collectively referred to as **folk psychology**. Although much of our folk psychology is probably reasonably accurate, it is clear that much of it is not. For example, most people believe that anger can be relieved by “letting it out”—perhaps by punching something or screaming loudly. Scientific research, however, has shown that this approach tends to leave people feeling more angry, not less (Bushman, 2002)¹. Likewise, most people believe that no one would confess to a crime that they had not committed unless perhaps that person was being physically tortured. But again, extensive empirical research has shown that false confessions are surprisingly common and occur for a variety of reasons (Kassin & Gudjonsson, 2004)².

Some Great Myths

In *50 Great Myths of Popular Psychology*, psychologist Scott Lilienfeld and colleagues discuss several widely held commonsense beliefs about human behavior that scientific research has shown to be incorrect (Lilienfeld, Lynn, Ruscio, & Beyerstein, 2010)³. Here is a short list:

- “People use only 10% of their brain power.”
- “Most people experience a midlife crisis in their 40's or 50's.”
- “Students learn best when teaching styles are matched to their learning styles.”
- “Low self-esteem is a major cause of psychological problems.”

- “Psychiatric admissions and crimes increase during full moons.”

How Could We Be So Wrong?

How can so many of our intuitive beliefs about human behavior be so wrong? Notice that this is an empirical question, and it just so happens that psychologists have conducted scientific research on it and identified many contributing factors (Gilovich, 1991)⁴. One is that forming detailed and accurate beliefs requires powers of observation, memory, and analysis to an extent that we do not naturally possess. It would be nearly impossible to count the number of words spoken by the women and men we happen to encounter, estimate the number of words they spoke per day, average these numbers for both groups, and compare them—all in our heads. This is why we tend to rely on mental shortcuts (what psychologists refer to as **heuristics**) in forming and maintaining our beliefs. For example, if a belief is widely shared—especially if it is endorsed by “experts”—and it makes intuitive sense, we tend to assume it is true. This is compounded by the fact that we then tend to focus on cases that confirm our intuitive beliefs and not on cases that dis-confirm them. This is called **confirmation bias**. For example, once we begin to believe that women are more talkative than men, we tend to notice and remember talkative women and silent men but ignore or forget silent women and talkative men. We also hold incorrect beliefs in part because it would be nice if they *were* true. For example, many people believe that calorie-reducing diets are an effective long-term treatment for obesity, yet a thorough review of the scientific evidence has shown that they are not (Mann et al., 2007)⁵. People may continue to believe in the effectiveness of dieting in part because it gives them hope for losing weight if they are obese or makes them feel good about their own “self-control” if they are not.

Scientists—especially psychologists—understand that they are just as susceptible as anyone else to intuitive but incorrect beliefs. This is why they cultivate an attitude of **skepticism**. Being skeptical does not mean being cynical or distrustful, nor does it mean questioning every belief or claim one comes across (which would be impossible anyway). Instead, it means pausing to consider alternatives and to search for evidence—especially systematically collected empirical evidence—when there is enough at stake to justify doing so. For example, imagine that you read a magazine article that claims that giving children a weekly allowance is a good way to help them develop financial responsibility. This is an interesting and potentially important claim (especially if you have children of your own). Taking an attitude of skepticism, however, would mean pausing to ask whether it might be instead that receiving an allowance merely teaches children to spend money—perhaps even to be more materialistic. Taking an attitude of skepticism would also mean asking what evidence supports the original claim. Is the author a scientific researcher? Is any scientific evidence cited? If the issue was important enough, it might also mean turning to the research literature to see if anyone else had studied it.

Because there is often not enough evidence to fully evaluate a belief or claim, scientists also cultivate a **tolerance for uncertainty**. They accept that there are many things that they simply do not know. For example, it turns out that there is no scientific evidence that receiving an allowance causes children to be

more financially responsible, nor is there any scientific evidence that it causes them to be materialistic. Although this kind of uncertainty can be problematic from a practical perspective—for example, making it difficult to decide what to do when our children ask for an allowance—it is exciting from a scientific perspective. If we do not know the answer to an interesting and empirically testable question, science, and perhaps even you as a researcher, may be able to provide the answer.

Notes

1. Bushman, B. J. (2002). Does venting anger feed or extinguish the flame? Catharsis, rumination, distraction, anger, and aggressive responding. *Personality and Social Psychology Bulletin*, 28, 724–731.
2. Kassin, S. M., & Gudjonsson, G. H. (2004). The psychology of confession evidence: A review of the literature and issues. *Psychological Science in the Public Interest*, 5, 33–67.
3. Lilienfeld, S. O., Lynn, S. J., Ruscio, J., & Beyerstein, B. L. (2010). *50 great myths of popular psychology*. Malden, MA: Wiley-Blackwell.
4. Gilovich, T. (1991). *How we know what isn't so: The fallibility of human reason in everyday life*. New York, NY: Free Press.
5. Mann, T., Tomiyama, A. J., Westling, E., Lew, A., Samuels, B., & Chatman, J. (2007). Medicare's search for effective obesity treatments: Diets are not the answer. *American Psychologist*, 62, 220–233.

5. Experimental and Clinical Psychologists

Learning Objectives

1. Define the clinical practice of psychology and distinguish it from experimental psychology.
2. Explain how science is relevant to clinical practice.
3. Define the concept of an empirically supported treatment and give some examples.

Who Conducts Scientific Research in Psychology?

Experimental Psychologists

Scientific research in psychology is generally conducted by people with doctoral degrees (usually the **doctor of philosophy [Ph.D.]**) and master's degrees in psychology and related fields, often supported by research assistants with bachelor's degrees or other relevant training. Some of them work for government agencies (e.g., doing research on the impact of public policies), national associations (e.g., the American Psychological Association), non-profit organizations (e.g., National Alliance on Mental Illness), or in the private sector (e.g., in product marketing and development; organizational behavior). However, the majority of them are college and university faculty, who often collaborate with their graduate and undergraduate students. Although some researchers are trained and licensed as clinicians for mental health work—especially those who conduct research in clinical psychology—the majority are not. Instead, they have expertise in one or more of the many other subfields of psychology: behavioral neuroscience, cognitive psychology, developmental psychology, personality psychology, social psychology, and so on. Doctoral-level researchers might be employed to conduct research full-time or, like many college and university faculty members, to conduct research in addition to teaching classes and serving their institution and community in other ways.

Of course, people also conduct research in psychology because they enjoy the intellectual and technical challenges involved and the satisfaction of contributing to scientific knowledge of human behavior. You might find that you enjoy the process too. If so, your college or university might offer opportunities to get involved in ongoing research as either a research assistant or a participant. Of course, you might find that you do not enjoy the process of conducting scientific research in psychology. But at least you will have a better understanding of where scientific knowledge in psychology comes from, an appreciation of its strengths and limitations, and an awareness of how it can be applied to solve practical problems in psychology and everyday life.

Scientific Psychology Blogs

A fun and easy way to follow current scientific research in psychology is to read any of the many excellent blogs devoted to summarizing and commenting on new findings. Among them are the following:

Research Digest, <http://digest.bps.org.uk/>

Talk Psych, <http://www.talkpsych.com/>

Brain Blogger, <http://brainblogger.com/>

Mind Hacks, <http://mindhacks.com/>

PsyBlog, <http://www.spring.org.uk>

You can also browse to <http://www.researchblogging.org>, select psychology as your topic, and read entries from a wide variety of blogs.

Clinical Psychologists

Psychology is the scientific study of behavior and mental processes. But it is also the application of scientific research to “help people, organizations, and communities function better” (American Psychological Association, 2011)¹. By far the most common and widely known application is the **clinical practice of psychology**—the diagnosis and treatment of psychological disorders and related problems. Let us use the term *clinical practice* broadly to refer to the activities of clinical and counseling psychologists, school psychologists, marriage and family therapists, licensed clinical social workers, and others who work with people individually or in small groups to identify and help address their psychological problems. It is important to consider the relationship between scientific research and clinical practice because many students are especially interested in clinical practice, perhaps even as a career.

The main point is that psychological disorders and other behavioral problems are part of the natural world. This means that questions about their nature, causes, and consequences are empirically testable and therefore subject to scientific study. As with other questions about human behavior, we cannot rely on our intuition or common sense for detailed and accurate answers. Consider, for example, that dozens of popular books and thousands of websites claim that adult children of alcoholics have a distinct personality profile, including low self-esteem, feelings of powerlessness, and difficulties with intimacy. Although this sounds plausible, scientific research has demonstrated that adult children of alcoholics are no more likely to have these problems than anybody else (Lilienfeld et al., 2010)². Similarly, questions about whether a particular psychotherapy is effective are empirically testable questions that can be answered by scientific research. If a new psychotherapy is an effective treatment for depression, then systematic observation should reveal that depressed people who receive this psychotherapy improve more than a similar group of depressed people

who do not receive this psychotherapy (or who receive some alternative treatment). Treatments that have been shown to work in this way are called **empirically supported treatments**.

Empirically Supported Treatments

An empirically supported treatment is one that has been studied scientifically and shown to result in greater improvement than no treatment, a placebo, or some alternative treatment. These include many forms of psychotherapy, which can be as effective as standard drug therapies. Among the forms of psychotherapy with strong empirical support are the following:

- **Acceptance and commitment therapy (ACT)**. for depression, mixed anxiety disorders, psychosis, chronic pain, and obsessive-compulsive disorder.
- **Behavioral couples therapy**. For alcohol use disorders.
- **Cognitive behavioral therapy (CBT)**. For many disorders including eating disorders, depression, anxiety disorders, etc.
- **Exposure therapy**. For post-traumatic stress disorder and phobias.
- **Exposure therapy with response prevention**. For obsessive-compulsive disorder.
- **Family-based treatment**. For eating disorders.

For a more complete list, see the following website, which is maintained by Division 12 of the American Psychological Association, the Society for Clinical Psychology: <http://www.div12.org/psychological-treatments>

Many in the clinical psychology community have argued that their field has not paid enough attention to scientific research—for example, by failing to use empirically supported treatments—and have suggested a variety of changes in the way clinicians are trained and treatments are evaluated and put into practice. Others believe that these claims are exaggerated and the suggested changes are unnecessary (Norcross, Beutler, & Levant, 2005)³. On both sides of the debate, however, there is agreement that a scientific approach to clinical psychology is essential if the goal is to diagnose and treat psychological problems based on detailed and accurate knowledge about those problems and the most effective treatments for them. So not only is it important for scientific research in clinical psychology to continue, but it is also important for clinicians who never conduct a scientific study themselves to be scientifically literate so that they can read and evaluate new research and make treatment decisions based on the best available evidence.

Notes

1. American Psychological Association. (2011). *About APA*. Retrieved from <http://www.apa.org/about>

2. Lilienfeld, S. O., Lynn, S. J., Ruscio, J., & Beyerstein, B. L. (2010). *50 great myths of popular psychology*. Malden, MA: Wiley-Blackwell.
3. Norcross, J. C., Beutler, L. E., & Levant, R. F. (Eds.). (2005). *Evidence-based practices in mental health: Debate and dialogue on the fundamental questions*. Washington, DC: American Psychological Association.

6. Key Takeaways and Exercises

Key Takeaways

- Knowledge is acquired in many ways including intuition, authority, rationalism, empiricism, and the scientific method
- Science is a general way of understanding the natural world. Its three fundamental features are systematic empiricism, empirical questions, and public knowledge.
- Psychology is a science because it takes the scientific approach to understanding human behavior.
- Pseudoscience refers to beliefs and activities that are claimed to be scientific but lack one or more of the three features of science. It is important to distinguish the scientific approach to understanding human behavior from the many pseudoscientific approaches.
- Psychologists conduct research in order to describe basic phenomenon, to make predictions about future behaviors, and to explain the causes of behavior.
- Basic research is conducted to learn about human behavior for its own sake, and applied research is conducted to solve some practical problem. Both are valuable, and the distinction between the two is not always clear-cut.
- People's intuitions about human behavior, also known as folk psychology, often turn out to be wrong. This is one primary reason that psychology relies on science rather than common sense.
- Researchers in psychology cultivate certain critical-thinking attitudes. One is skepticism. They search for evidence and consider alternatives before accepting a claim about human behavior as true. Another is tolerance for uncertainty. They withhold judgment about whether a claim is true or not when there is insufficient evidence to decide.
- Scientific research in psychology is conducted mainly by people with doctoral degrees in psychology and related fields, most of whom are college and university faculty members. They do so for professional and for personal reasons, as well as to contribute to scientific knowledge about human behavior. Most psychologists are experimental psychologists and they conduct research.
- The clinical practice of psychology—the diagnosis and treatment of psychological problems—is one important application of the scientific discipline of psychology.
- Scientific research is relevant to clinical practice because it provides detailed and accurate knowledge about psychological problems and establishes whether treatments are effective.

Exercises

- Practice: Consider three things you know and determine how you acquired that knowledge (authority, intuition, rationalism, empiricism, the scientific method).
- Practice: Try to generate different research questions to describe, predict, and explain a phenomenon that interests you.

- Practice: Based on your own experience or on things you have already learned about psychology, list three basic research questions and three applied research questions of interest to you.
- Practice: List three empirical questions about human behavior. List three nonempirical questions about human behavior.
- Practice: For each of the following intuitive beliefs about human behavior, list three reasons that it might be true and three reasons that it might not be true:
 - You cannot truly love another person unless you love yourself.
 - People who receive “crisis counseling” immediately after experiencing a traumatic event are better able to cope with that trauma in the long term.
 - Studying is most effective when it is always done in the same location.
- Watch the following video, in which psychologist Scott Lilienfeld talks about confirmation bias, tunnel vision, and using evidence to evaluate the world around us:



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://kpu.pressbooks.pub/psychmethods4e/?p=250#oembed-2>

- Reading in print? Go to <https://youtu.be/gKVe0-pWjA0> or scan this QR code with your phone:



- Discussion: Consider the following psychological claim. “People’s choice of spouse is strongly influenced by their perception of their own parents. Some choose a spouse who is similar in some way to one of their parents. Others choose a spouse who is different from one of their parents.” Is this claim falsifiable? Why or why not?
- Discussion: People sometimes suggest that psychology cannot be a science because either (a) human behavior cannot be predicted with perfect accuracy or (b) much of its subject matter (e.g., thoughts and feelings) cannot be observed directly. Do you agree or disagree with each of these ideas? Why?
- Watch the following video by PHD Comics for an overview of open access publishing and why it matters:



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://kpu.pressbooks.pub/psychmethods4e/?p=250#oembed-1>

- Reading in print? Go to <https://youtu.be/L5rVH1KGBCY> or scan this QR code with your phone:



- Discussion: Some clinicians argue that what they do is an “art form” based on intuition and personal experience and therefore cannot be evaluated scientifically. Write a paragraph about how satisfied you would be with such a clinician and why from each of three perspectives:
 - a potential client of the clinician
 - a judge who must decide whether to allow the clinician to testify as an expert witness in a child abuse case
 - an insurance company representative who must decide whether to reimburse the clinician for their services
- Practice: Create a short list of questions that a client could ask a clinician to determine whether they pay sufficient attention to scientific research.

Video Attributions

- [ESC 2017 – Scott Lilienfeld: “Tunnel Vision: Confirmation Bias”](#) © [ESC European Skeptics Congress](#) is licensed under a [CC BY \(Attribution\)](#) license
- [“Open Access Explained!”](#) by [Piled Higher and Deeper \(PHD Comics\)](#). [CC BY \(Attribution\)](#)

CHAPTER II

OVERVIEW OF THE SCIENTIFIC METHOD

Here is the abstract of a 2014 article in the journal *Psychological Science*.

Taking notes on laptops rather than in longhand is increasingly common. Many researchers have suggested that laptop note taking is less effective than longhand note taking for learning. Prior studies have primarily focused on students' capacity for multitasking and distraction when using laptops. The present research suggests that even when laptops are used solely to take notes, they may still be impairing learning because their use results in shallower processing. In three studies, we found that students who took notes on laptops performed worse on conceptual questions than students who took notes longhand. We show that whereas taking more notes can be beneficial, laptop note takers' tendency to transcribe lectures verbatim rather than processing information and reframing it in their own words is detrimental to learning. (Mueller & Oppenheimer, 2014, p. 1159)¹

In this abstract, the researcher has identified a research question—about the effect of taking notes on a laptop on learning—and identified why it is worthy of investigation—because the practice is ubiquitous and may be harmful to learning. In this chapter, we give you a broad overview of the various stages of the research process. These include finding a topic of investigation, reviewing the literature, refining your research question and generating a hypothesis, designing and conducting a study, analyzing the data, coming to conclusions, and reporting the results.

Notes

1. Mueller, P. A., & Oppenheimer, D. M. (2014). The pen is mightier than the keyboard: Advantages of longhand over laptop note taking. *Psychological Science*, 25(6), 1159–1168.

7. A Model of Scientific Research in Psychology

Learning Objectives

1. Review a general model of scientific research in psychology.

Figure 2.1 presents a simple model of scientific research in psychology. The researchers formulate a research question, conduct an empirical study designed to answer the question, analyze the resulting data, draw conclusions about the answer to the question, and publishes the results so that they become part of the research literature (i.e., all the published research in that field). Because the research literature is one of the primary sources of new research questions, this process can be thought of as a cycle. New research leads to new questions, which lead to new research, and so on. Figure 2.1 also indicates that research questions can originate outside of this cycle either with informal observations or with practical problems that need to be solved. But even in these cases, the researcher would start by checking the research literature to see if the question had already been answered and to refine it based on what previous research had already found.

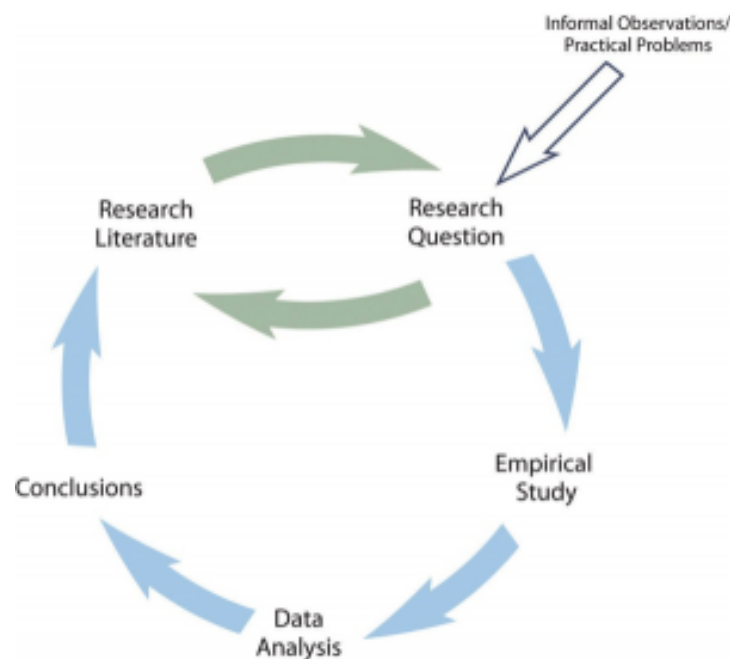


Figure 2.1 A Simple Model of Scientific Research in Psychology

The research by Mehl and his colleagues is described nicely by this model. Their research question—whether women are more talkative than men—was suggested to them both by people’s stereotypes and by claims published in the research literature about the relative talkativeness of women and men. When they checked the research literature, however, they found that this question had not been adequately addressed in scientific studies. They then conducted a careful empirical study, analyzed the results (finding very little difference between women and men), formed their conclusions, and published their work so that it became part of the research literature. The publication of their article is not the end of the story, however, because their work suggests many new questions (about the reliability of the result, about potential cultural differences, etc.) that will likely be taken up by them and by other researchers inspired by their work.



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As another example, consider that as cell phones became more widespread during the 1990s, people began to wonder whether, and to what extent, cell phone use had a negative effect on driving. Many psychologists decided to tackle this question scientifically (e.g., Collet, Guillot, & Petit, 2010)¹. It was clear from previously published research that engaging in a simple verbal task impairs performance on a perceptual or motor task carried out at the same time, but no one had studied the effect specifically of cell phone use on driving. Under carefully controlled conditions, these researchers compared people’s driving performance while using a cell phone with their performance while not using a cell phone, both in the lab and on the road. They found that people’s ability to detect road hazards, reaction time, and maintain control of the vehicle were all impaired by cell phone use. Each new study was published and became part of the growing research literature on this topic. For instance, other research teams subsequently demonstrated that cell phone conversations carry a greater risk than conversations with a passenger who is aware of driving conditions, which often become a point of conversation (e.g., Drews, Pasupathi, & Strayer, 2004)².



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Video Attributions

- “[Understanding driver distraction](#)” by [American Psychological Association](#). Standard YouTube Licence.

Notes

1. Collet, C., Guillot, A., & Petit, C. (2010). Phoning while driving I: A review of epidemiological, psychological, behavioral and physiological studies. *Ergonomics*, 53, 589–601.
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8. Finding a Research Topic

Learning Objectives

1. Learn some common sources of research ideas.
2. Define the research literature in psychology and give examples of sources that are part of the research literature and sources that are not.
3. Describe and use several methods for finding previous research on a particular research idea or question.

Good research must begin with a good research question. Yet coming up with good research questions is something that novice researchers often find difficult and stressful. One reason is that this is a creative process that can appear mysterious—even magical—with experienced researchers seeming to pull interesting research questions out of thin air. However, psychological research on creativity has shown that it is neither as mysterious nor as magical as it appears. It is largely the product of ordinary thinking strategies and persistence (Weisberg, 1993)¹. This section covers some fairly simple strategies for finding general research ideas, turning those ideas into empirically testable research questions, and finally evaluating those questions in terms of how interesting they are and how feasible they would be to answer.

Finding Inspiration

Research questions often begin as more general research ideas—usually focusing on some behavior or psychological characteristic: talkativeness, learning, depression, bungee jumping, and so on. Before looking at how to turn such ideas into empirically testable research questions, it is worth looking at where such ideas come from in the first place. Three of the most common sources of inspiration are informal observations, practical problems, and previous research.

Informal observations include direct observations of our own and others' behavior as well as secondhand observations from non-scientific sources such as newspapers, books, blogs, and so on. For example, you might notice that you always seem to be in the slowest moving line at the grocery store. Could it be that most people think the same thing? Or you might read in a local newspaper about people donating money and food to a local family whose house has burned down and begin to wonder about who makes such donations and why. Some of the most famous research in psychology has been inspired by informal observations. Stanley Milgram's famous research on obedience to authority, for example, was inspired in part by journalistic reports of the trials of accused Nazi war criminals—many of whom claimed that they

were only obeying orders. This led him to wonder about the extent to which ordinary people will commit immoral acts simply because they are ordered to do so by an authority figure (Milgram, 1963)².

Practical problems can also inspire research ideas, leading directly to applied research in such domains as law, health, education, and sports. Does taking lecture notes by hand improve students' exam performance? How effective is psychotherapy for depression compared to drug therapy? To what extent do cell phones impair people's driving ability? How can we teach children to read more efficiently? What is the best mental preparation for running a marathon?

Probably the most common inspiration for new research ideas, however, is previous research. Recall that science is a kind of large-scale collaboration in which many different researchers read and evaluate each other's work and conduct new studies to build on it. Of course, experienced researchers are familiar with previous research in their area of expertise and probably have a long list of ideas. This suggests that novice researchers can find inspiration by consulting with a more experienced researcher (e.g., students can consult a faculty member). But they can also find inspiration by picking up a copy of almost any professional journal and reading the titles and abstracts. In one typical issue of *Psychological Science*, for example, you can find articles on the perception of shapes, anti-Semitism, police lineups, the meaning of death, second-language learning, people who seek negative emotional experiences, and many other topics. If you can narrow your interests down to a particular topic (e.g., memory) or domain (e.g., health care), you can also look through more specific journals, such as *Memory & Cognition* or *Health Psychology*.



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Reviewing the Research Literature

Once again, one of the most common sources of inspiration is previous research. Therefore, it is important to review the literature early in the research process. The **research literature** in any field is all the published research in that field. Reviewing the research literature means finding, reading, and summarizing the published research relevant to your topic of interest. In addition to helping you discover new research questions, reviewing the literature early in the research process can help you in several other ways.

- It can tell you if a research question has already been answered.
- It can help you evaluate the interestingness of a research question.
- It can give you ideas for how to conduct your own study.
- It can tell you how your study fits into the research literature.



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The research literature in psychology is enormous—including millions of scholarly articles and books dating to the beginning of the field—and it continues to grow. Although its boundaries are somewhat fuzzy, the research literature definitely does not include self-help and other pop psychology books, dictionary and encyclopedia entries, websites, and similar sources that are intended mainly for the general public. These are considered unreliable because they are not reviewed by other researchers and are often based on little more than common sense or personal experience. Wikipedia contains much valuable information, but because its authors are anonymous and may not have any formal training or expertise in that subject area, and its content continually changes it is unsuitable as a basis of sound scientific research. For our purposes, it helps to define the research literature as consisting almost entirely of two types of sources: articles in professional journals, and scholarly books in psychology and related fields.

Professional Journals

Professional journals are periodicals that publish original research articles. There are thousands of professional journals that publish research in psychology and related fields. They are usually published monthly or quarterly in individual issues, each of which contains several articles. The issues are organized into volumes, which usually consist of all the issues for a calendar year. Some journals are published in hard copy only, others in both hard copy and electronic form, and still others in electronic form only.

Most articles in professional journals are one of two basic types: empirical research reports and review articles. **Empirical research reports** describe one or more new empirical studies conducted by the authors. They introduce a research question, explain why it is interesting, review previous research, describe their method and results, and draw their conclusions. **Review articles** summarize previously published research on a topic and usually present new ways to organize or explain the results. When a review article is devoted

primarily to presenting a new theory, it is often referred to as a **theoretical article**. When a review article provides a statistical summary of all of the previous results it is referred to as a **meta-analysis**.



Figure 2.2 Small Sample of the Thousands of Professional Journals That Publish Research in Psychology and Related Fields

Most professional journals in psychology undergo a process of **double-blind peer review**. Researchers who want to publish their work in the journal submit a manuscript to the editor—who is generally an established researcher too—who in turn sends it to two or three experts on the topic. Each reviewer reads the manuscript, writes a critical but constructive review, and sends the review back to the editor along with recommendations about whether the manuscript should be published or not. The editor then decides whether to accept the article for publication, ask the authors to make changes and resubmit it for further consideration, or reject it outright. In any case, the editor forwards the reviewers' written comments to the researchers so that they can revise their manuscript accordingly. This entire process is double-blind, as the reviewers do not know the identity of the researcher(s) and vice versa. Double-blind peer review is helpful because it ensures that the work meets basic standards of the field before it can enter the research literature. However, in order to increase transparency and accountability, some newer open access journals

(e.g., [Frontiers in Psychology](#)) utilize an open peer review process wherein the identities of the reviewers (which remain concealed during the peer review process) are published alongside the journal article.

Scholarly Books

Scholarly books are books written by researchers and practitioners mainly for use by other researchers and practitioners. A **monograph** is written by a single author or a small group of authors and usually, gives a coherent presentation of a topic much like an extended review article. **Edited volumes** have an editor or a small group of editors who recruit many authors to write separate chapters on different aspects of the same topic. Although edited volumes can also give a coherent presentation of the topic, it is not unusual for each chapter to take a different perspective or even for the authors of different chapters to openly disagree with each other. In general, scholarly books undergo a peer review process similar to that used by professional journals.

Literature Search Strategies

Using PsycINFO and Other Databases

The primary method used to search the research literature involves using one or more electronic databases. These include Academic Search Premier, JSTOR, and ProQuest for all academic disciplines, ERIC for education, and PubMed for medicine and related fields. The most important for our purposes, however, is PsycINFO, which is produced by the American Psychological Association (APA). **PsycINFO** is so comprehensive—covering thousands of professional journals and scholarly books going back more than 100 years—that for most purposes its content is synonymous with the research literature in psychology. Like most such databases, PsycINFO is usually available through your university library.

PsycINFO consists of individual records for each article, book chapter, or book in the database. Each record includes basic publication information, an abstract or summary of the work (like the one presented at the start of this chapter), and a list of other works cited by that work. A computer interface allows entering one or more search terms and returns any records that contain those search terms. (These interfaces are provided by different vendors and therefore can look somewhat different depending on the library you use.) Each record also contains lists of keywords that describe the content of the work and also a list of index terms. The index terms are especially helpful because they are standardized. Research on differences between females and males, for example, is always indexed under “Human Sex Differences.” Research on note-taking is always indexed under the term “Learning Strategies.” If you do not know the appropriate index terms, PsycINFO includes a thesaurus that can help you find them.

Given that there are nearly four million records in PsycINFO, you may have to try a variety of search terms in different combinations and at different levels of specificity before you find what you are looking for.

Imagine, for example, that you are interested in the question of whether males and females differ in terms of their ability to recall experiences from when they were very young. If you were to enter the search term “memory,” it would return far too many records to look through individually. This is where the thesaurus helps. Entering “memory” into the thesaurus provides several more specific index terms—one of which is “early memories.” While searching for “early memories” among the index terms still returns too many to look through individually—combining it with “human sex differences” as a second search term returns fewer articles, many of which are highly relevant to the topic.

Depending on the vendor that provides the interface to PsycINFO, you may be able to save, print, or e-mail the relevant PsycINFO records. The records might even contain links to full-text copies of the works themselves. (PsycARTICLES is a database that provides full-text access to articles in all journals published by the APA.) If not, and you want a copy of the work, you will have to find out if your library carries the journal or has the book and the hard copy on the library shelves. Be sure to ask a librarian if you need help.

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Using Other Search Techniques

In addition to entering search terms into PsycINFO and other databases, there are several other techniques you can use to search the research literature. First, if you have one good article or book chapter on your topic—a recent review article is best—you can look through the reference list of that article for other relevant articles, books, and book chapters. In fact, you should do this with any relevant article or book chapter you find. You can also start with a classic article or book chapter on your topic, find its record in PsycINFO (by entering the author’s name or article’s title as a search term), and link from there to a list of other works in PsycINFO that cite that classic article. This works because other researchers working on your topic are likely to be aware of the classic article and cite it in their own work. You can also do a general Internet search using search terms related to your topic or the name of a researcher who conducts research on your topic. This might lead you directly to works that are part of the research literature (e.g., articles in open-access journals or posted on researchers’ own websites). The search engine Google Scholar is especially useful for this purpose. A general Internet search might also lead you to websites that are not part of the research literature but might provide references to works that are. Finally, you can talk to people (e.g., your instructor or other faculty members in psychology) who know something about your topic and can suggest relevant articles and book chapters.





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What to Search For

When you do a literature review, you need to be selective. Not every article, book chapter, and book that relates to your research idea or question will be worth obtaining, reading, and integrating into your review. Instead, you want to focus on sources that help you do four basic things: (a) refine your research question, (b) identify appropriate research methods, (c) place your research in the context of previous research, and (d) write an effective research report. Several basic principles can help you find the most useful sources.

First, it is best to focus on recent research, keeping in mind that what counts as recent depends on the topic. For newer topics that are actively being studied, “recent” might mean published in the past year or two. For older topics that are receiving less attention right now, “recent” might mean within the past 10 years. You will get a feel for what counts as recent for your topic when you start your literature search. A good general rule, however, is to start with sources published in the past five years. The main exception to this rule would be classic articles that turn up in the reference list of nearly every other source. If other researchers think that this work is important, even though it is old, then, by all means, you should include it in your review.

Second, you should look for review articles on your topic because they will provide a useful overview of it—often discussing important definitions, results, theories, trends, and controversies—giving you a good sense of where your own research fits into the literature. You should also look for empirical research reports addressing your question or similar questions, which can give you ideas about how to measure your variables and collect your data. As a general rule, it is good to use methods that others have already used successfully unless you have good reasons not to. Finally, you should look for sources that provide information that can help you argue for the interestingness of your research question. For a study on the effects of cell phone use on driving ability, for example, you might look for information about how widespread cell phone use is, how frequent and costly motor vehicle crashes are, and so on.

How many sources are enough for your literature review? This is a difficult question because it depends on how extensively your topic has been studied and also on your own goals. One study found that across a variety of professional journals in psychology, the average number of sources cited per article was about 50 (Adair & Vohra, 2003)³. This gives a rough idea of what professional researchers consider to be adequate. As a student, you might be assigned a much lower minimum number of references to include, but the principles for selecting the most useful ones remain the same.



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- “[Sample PsycINFO Search on EBSCOhost](#)” by [APA Publishing Training](#). Standard YouTube Licence.
- “[Using Google Scholar \(CLIP\)](#)” by [clipinfolit](#). [CC BY \(Attribution\)](#)

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1. Weisberg, R. W. (1993). *Creativity: Beyond the myth of genius*. New York, NY: Freeman.
2. Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology*, 67, 371–378.
3. Adair, J. G., & Vohra, N. (2003). The explosion of knowledge, references, and citations: Psychology’s unique response to a crisis. *American Psychologist*, 58, 15–23.

9. Generating Good Research Questions

Learning Objectives

1. Describe some techniques for turning research ideas into empirical research questions and use those techniques to generate questions.
2. Explain what makes a research question interesting and evaluate research questions in terms of their interestingness.

Generating Empirically Testable Research Questions

Once you have a research idea, you need to use it to generate one or more empirically testable research questions, that is, questions expressed in terms of a single variable or relationship between variables. One way to do this is to look closely at the discussion section in a recent research article on the topic. This is the last major section of the article, in which the researchers summarize their results, interpret them in the context of past research, and suggest directions for future research. These suggestions often take the form of specific research questions, which you can then try to answer with additional research. This can be a good strategy because it is likely that the suggested questions have already been identified as interesting and important by experienced researchers.

But you may also want to generate your own research questions. How can you do this? First, if you have a particular behavior or psychological characteristic in mind, you can simply conceptualize it as a variable and ask how frequent or intense it is. How many words on average do people speak per day? How accurate are our memories of traumatic events? What percentage of people have sought professional help for depression? If the question has never been studied scientifically—which is something that you will learn when you conduct your literature review—then it might be interesting and worth pursuing.

If scientific research has already answered the question of how frequent or intense the behavior or characteristic is, then you should consider turning it into a question about a relationship between that behavior or characteristic and some other variable. One way to do this is to ask yourself the following series of more general questions and write down all the answers you can think of.

- What are some possible causes of the behavior or characteristic?
- What are some possible effects of the behavior or characteristic?
- What types of people might exhibit more or less of the behavior or characteristic?
- What types of situations might elicit more or less of the behavior or characteristic?

In general, each answer you write down can be conceptualized as a second variable, suggesting a question about a relationship. If you were interested in talkativeness, for example, it might occur to you that a possible cause of this psychological characteristic is family size. Is there a relationship between family size and talkativeness? Or it might occur to you that people seem to be more talkative in same-sex groups than mixed-sex groups. Is there a difference in the average level of talkativeness of people in same-sex groups and people in mixed-sex groups? This approach should allow you to generate many different empirically testable questions about almost any behavior or psychological characteristic.

If through this process you generate a question that has never been studied scientifically—which again is something that you will learn in your literature review—then it might be interesting and worth pursuing. But what if you find that it has been studied scientifically? Although novice researchers often want to give up and move on to a new question at this point, this is not necessarily a good strategy. For one thing, the fact that the question has been studied scientifically and the research published suggests that it is of interest to the scientific community. For another, the question can almost certainly be refined so that its answer will still contribute something new to the research literature. Again, asking yourself a series of more general questions about the relationship is a good strategy.

- Are there other ways to define and measure the variables?
- Are there types of people for whom the relationship might be stronger or weaker?
- Are there situations in which the relationship might be stronger or weaker—including situations with practical importance?

For example, research has shown that women and men speak about the same number of words per day—but this was when talkativeness was measured in terms of the number of words spoken per day among university students in the United States and Mexico. We can still ask whether other ways of measuring talkativeness—perhaps the number of different people spoken to each day—produce the same result. Or we can ask whether studying elderly people or people from other cultures produces the same result. Again, this approach should help you generate many different research questions about almost any relationship.

Evaluating Research Questions

Researchers usually generate many more research questions than they ever attempt to answer. This means they must have some way of evaluating the research questions they generate so that they can choose which ones to pursue. In this section, we consider two criteria for evaluating research questions: the interestingness of the question and the feasibility of answering it.

Interestingness

How often do people tie their shoes? Do people feel pain when you punch them in the jaw? Are women more likely to wear makeup than men? Do people prefer vanilla or chocolate ice cream? Although it would

be a fairly simple matter to design a study and collect data to answer these questions, you probably would not want to because they are not interesting. We are not talking here about whether a research question is interesting to us personally but whether it is interesting to people more generally and, especially, to the scientific community. But what makes a research question interesting in this sense? Here we look at three factors that affect the **interestingness** of a research question: the answer is in doubt, the answer fills a gap in the research literature, and the answer has important practical implications.

First, a research question is interesting to the extent that its answer is in doubt. Obviously, questions that have been answered by scientific research are no longer interesting as the subject of new empirical research. But the fact that a question has not been answered by scientific research does not necessarily make it interesting. There has to be some reasonable chance that the answer to the question will be something that we did not already know. But how can you assess this before actually collecting data? One approach is to try to think of reasons to expect different answers to the question—especially ones that seem to conflict with common sense. If you can think of reasons to expect at least two different answers, then the question might be interesting. If you can think of reasons to expect only one answer, then it probably is not. The question of whether women are more talkative than men is interesting because there are reasons to expect both answers. The existence of the stereotype itself suggests the answer could be yes, but the fact that women's and men's verbal abilities are fairly similar suggests the answer could be no. The question of whether people feel pain when you punch them in the jaw is not interesting because there is absolutely no reason to think that the answer could be anything other than a resounding yes.

A second important factor to consider when deciding if a research question is interesting is whether answering it will fill a gap in the research literature. Again, this means in part that the question has not already been answered by scientific research. But it also means that the question is in some sense a natural one for people who are familiar with the research literature. For example, the question of whether taking lecture notes by hand can help improve students' exam performance would be likely to occur to anyone who was familiar with research on note taking and the ineffectiveness of shallow processing on learning.

A final factor to consider when deciding whether a research question is interesting is whether its answer has important practical implications. Again, the question of whether taking notes by hand improves learning has important implications for education, including classroom policies concerning technology use. The question of whether cell phone use impairs driving is interesting because it is relevant to the personal safety of everyone who travels by car and to the debate over whether cell phone use should be restricted by law.

Feasibility

A second important criterion for evaluating research questions is the **feasibility** of successfully answering them. There are many factors that affect feasibility, including time, money, equipment and materials, technical knowledge and skill, and access to research participants. Clearly, researchers need to take these factors into account so that they do not waste time and effort pursuing research that they cannot complete successfully.

Looking through a sample of professional journals in psychology will reveal many studies that are

complicated and difficult to carry out. These include longitudinal designs in which participants are tracked over many years, neuroimaging studies in which participants' brain activity is measured while they carry out various mental tasks, and complex non-experimental studies involving several variables and complicated statistical analyses. Keep in mind, though, that such research tends to be carried out by teams of highly trained researchers whose work is often supported in part by government and private grants. Also, keep in mind that research does not have to be complicated or difficult to produce interesting and important results. Looking through a sample of professional journals will also reveal studies that are relatively simple and easy to carry out—perhaps involving a convenience sample of university students and a paper-and-pencil task.

A final point here is that it is generally good practice to use methods that have already been used successfully by other researchers. For example, if you want to manipulate people's moods to make some of them happy, it would be a good idea to use one of the many approaches that have been used successfully by other researchers (e.g., paying them a compliment). This is good not only for the sake of feasibility—the approach is “tried and true”—but also because it provides greater continuity with previous research. This makes it easier to compare your results with those of other researchers and to understand the implications of their research for yours, and vice versa.

10. Developing a Hypothesis

Learning Objectives

1. Distinguish between a theory and a hypothesis.
2. Discover how theories are used to generate hypotheses and how the results of studies can be used to further inform theories.
3. Understand the characteristics of a good hypothesis.

Theories and Hypotheses

Before describing how to develop a hypothesis, it is important to distinguish between a theory and a hypothesis. A **theory** is a coherent explanation or interpretation of one or more phenomena. Although theories can take a variety of forms, one thing they have in common is that they go beyond the phenomena they explain by including variables, structures, processes, functions, or organizing principles that have not been observed directly. Consider, for example, Zajonc's theory of social facilitation and social inhibition (1965)¹. He proposed that being watched by others while performing a task creates a general state of physiological arousal, which increases the likelihood of the dominant (most likely) response. So for highly practiced tasks, being watched increases the tendency to make correct responses, but for relatively unpracticed tasks, being watched increases the tendency to make incorrect responses. Notice that this theory—which has come to be called drive theory—provides an explanation of both social facilitation and social inhibition that goes beyond the phenomena themselves by including concepts such as “arousal” and “dominant response,” along with processes such as the effect of arousal on the dominant response.

Outside of science, referring to an idea as a theory often implies that it is untested—perhaps no more than a wild guess. In science, however, the term theory has no such implication. A theory is simply an explanation or interpretation of a set of phenomena. It can be untested, but it can also be extensively tested, well supported, and accepted as an accurate description of the world by the scientific community. The theory of evolution by natural selection, for example, is a theory because it is an explanation of the diversity of life on earth—not because it is untested or unsupported by scientific research. On the contrary, the evidence for this theory is overwhelmingly positive and nearly all scientists accept its basic assumptions as accurate. Similarly, the “germ theory” of disease is a theory because it is an explanation of the origin of various diseases, not because there is any doubt that many diseases are caused by microorganisms that infect the body.

A **hypothesis**, on the other hand, is a specific prediction about a new phenomenon that should be observed if a particular theory is accurate. It is an explanation that relies on just a few key concepts. Hypotheses are

often specific predictions about what will happen in a particular study. They are developed by considering existing evidence and using reasoning to infer what will happen in the specific context of interest. Hypotheses are often but not always derived from theories. So a hypothesis is often a prediction based on a theory but some hypotheses are a-theoretical and only after a set of observations have been made, is a theory developed. This is because theories are broad in nature and they explain larger bodies of data. So if our research question is really original then we may need to collect some data and make some observations before we can develop a broader theory.

Theories and hypotheses always have this *if-then* relationship. “If drive theory is correct, *then* cockroaches should run through a straight runway faster, and a branching runway more slowly, when other cockroaches are present.” Although hypotheses are usually expressed as statements, they can always be rephrased as questions. “Do cockroaches run through a straight runway faster when other cockroaches are present?” Thus deriving hypotheses from theories is an excellent way of generating interesting research questions.

But how do researchers derive hypotheses from theories? One way is to generate a research question using the techniques discussed in this chapter and then ask whether any theory implies an answer to that question. For example, you might wonder whether expressive writing about positive experiences improves health as much as expressive writing about traumatic experiences. Although this question is an interesting one on its own, you might then ask whether the habituation theory—the idea that expressive writing causes people to habituate to negative thoughts and feelings—implies an answer. In this case, it seems clear that if the habituation theory is correct, then expressive writing about positive experiences should not be effective because it would not cause people to habituate to negative thoughts and feelings. A second way to derive hypotheses from theories is to focus on some component of the theory that has not yet been directly observed. For example, a researcher could focus on the process of habituation—perhaps hypothesizing that people should show fewer signs of emotional distress with each new writing session.

Among the very best hypotheses are those that distinguish between competing theories. For example, Norbert Schwarz and his colleagues considered two theories of how people make judgments about themselves, such as how assertive they are (Schwarz et al., 1991)². Both theories held that such judgments are based on relevant examples that people bring to mind. However, one theory was that people base their judgments on the *number* of examples they bring to mind and the other was that people base their judgments on how *easily* they bring those examples to mind. To test these theories, the researchers asked people to recall either six times when they were assertive (which is easy for most people) or 12 times (which is difficult for most people). Then they asked them to judge their own assertiveness. Note that the number-of-examples theory implies that people who recalled 12 examples should judge themselves to be more assertive because they recalled more examples, but the ease-of-examples theory implies that participants who recalled six examples should judge themselves as more assertive because recalling the examples was easier. Thus the two theories made opposite predictions so that only one of the predictions could be confirmed. The surprising result was that participants who recalled fewer examples judged themselves to be more assertive—providing particularly convincing evidence in favor of the ease-of-retrieval theory over the number-of-examples theory.

Theory Testing

The primary way that scientific researchers use theories is sometimes called the **hypothetico-deductive method** (although this term is much more likely to be used by philosophers of science than by scientists themselves). Researchers begin with a set of phenomena and either construct a theory to explain or interpret them or choose an existing theory to work with. They then make a prediction about some new phenomenon that should be observed if the theory is correct. Again, this prediction is called a hypothesis. The researchers then conduct an empirical study to test the hypothesis. Finally, they reevaluate the theory in light of the new results and revise it if necessary. This process is usually conceptualized as a cycle because the researchers can then derive a new hypothesis from the revised theory, conduct a new empirical study to test the hypothesis, and so on. As Figure 2.3 shows, this approach meshes nicely with the model of scientific research in psychology presented earlier in the textbook—creating a more detailed model of “theoretically motivated” or “theory-driven” research.

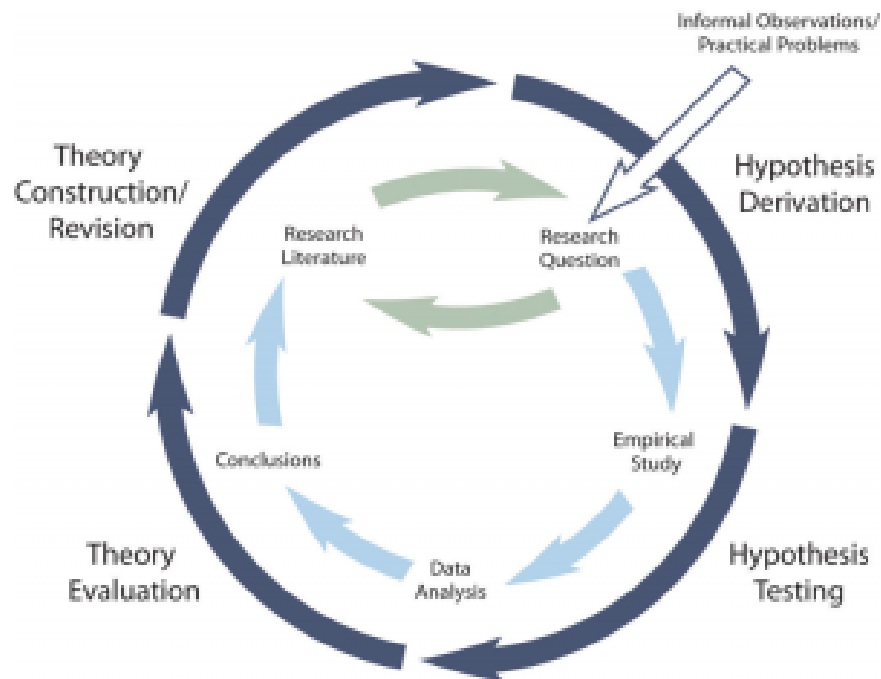


Figure 2.3 Hypothetico-Deductive Method Combined With the General Model of Scientific Research in Psychology Together they form a model of theoretically motivated research.

As an example, let us consider Zajonc’s research on social facilitation and inhibition. He started with a somewhat contradictory pattern of results from the research literature. He then constructed his drive theory, according to which being watched by others while performing a task causes physiological arousal, which increases an organism’s tendency to make the dominant response. This theory predicts social facilitation for well-learned tasks and social inhibition for poorly learned tasks. He now had a theory that organized previous results in a meaningful way—but he still needed to test it. He hypothesized that if

his theory was correct, he should observe that the presence of others improves performance in a simple laboratory task but inhibits performance in a difficult version of the very same laboratory task. To test this hypothesis, one of the studies he conducted used cockroaches as subjects (Zajonc, Heingartner, & Herman, 1969)³. The cockroaches ran either down a straight runway (an easy task for a cockroach) or through a cross-shaped maze (a difficult task for a cockroach) to escape into a dark chamber when a light was shined on them. They did this either while alone or in the presence of other cockroaches in clear plastic “audience boxes.” Zajonc found that cockroaches in the straight runway reached their goal more quickly in the presence of other cockroaches, but cockroaches in the cross-shaped maze reached their goal more slowly when they were in the presence of other cockroaches. Thus he confirmed his hypothesis and provided support for his drive theory. (Zajonc also showed that drive theory existed in humans [Zajonc & Sales, 1966]⁴ in many other studies afterward).

Incorporating Theory into Your Research

When you write your research report or plan your presentation, be aware that there are two basic ways that researchers usually include theory. The first is to raise a research question, answer that question by conducting a new study, and then offer one or more theories (usually more) to explain or interpret the results. This format works well for applied research questions and for research questions that existing theories do not address. The second way is to describe one or more existing theories, derive a hypothesis from one of those theories, test the hypothesis in a new study, and finally reevaluate the theory. This format works well when there is an existing theory that addresses the research question—especially if the resulting hypothesis is surprising or conflicts with a hypothesis derived from a different theory.

To use theories in your research will not only give you guidance in coming up with experiment ideas and possible projects, but it lends legitimacy to your work. Psychologists have been interested in a variety of human behaviors and have developed many theories along the way. Using established theories will help you break new ground as a researcher, not limit you from developing your own ideas.

Characteristics of a Good Hypothesis

There are three general characteristics of a good hypothesis. First, a good hypothesis must be **testable and falsifiable**. We must be able to test the hypothesis using the methods of science and if you’ll recall Popper’s falsifiability criterion, it must be possible to gather evidence that will disconfirm the hypothesis if it is indeed false. Second, a good hypothesis must be **logical**. As described above, hypotheses are more than just a random guess. Hypotheses should be informed by previous theories or observations and logical reasoning. Typically, we begin with a broad and general theory and use *deductive reasoning* to generate a more specific hypothesis to test based on that theory. Occasionally, however, when there is no theory to inform our hypothesis, we use *inductive reasoning* which involves using specific observations or research findings to form a more general hypothesis. Finally, the hypothesis should be **positive**. That is, the hypothesis should make a positive statement about the existence of a relationship or effect, rather than

a statement that a relationship or effect does not exist. As scientists, we don't set out to show that relationships do not exist or that effects do not occur so our hypotheses should not be worded in a way to suggest that an effect or relationship does not exist. The nature of science is to assume that something does not exist and then seek to find evidence to prove this wrong, to show that it really does exist. That may seem backward to you but that is the nature of the scientific method. The underlying reason for this is beyond the scope of this chapter but it has to do with statistical theory.

Notes

1. Zajonc, R. B. (1965). Social facilitation. *Science*, 149, 269–274
2. Schwarz, N., Bless, H., Strack, F., Klumpp, G., Rittenauer-Schatka, H., & Simons, A. (1991). Ease of retrieval as information: Another look at the availability heuristic. *Journal of Personality and Social Psychology*, 61, 195–202.
3. Zajonc, R. B., Heingartner, A., & Herman, E. M. (1969). Social enhancement and impairment of performance in the cockroach. *Journal of Personality and Social Psychology*, 13, 83–92.
4. Zajonc, R.B. & Sales, S.M. (1966). Social facilitation of dominant and subordinate responses. *Journal of Experimental Social Psychology*, 2, 160–168.

II. Designing a Research Study

Learning Objectives

1. Define the concept of a variable, distinguish quantitative from categorical variables, and give examples of variables that might be of interest to psychologists.
2. Explain the difference between a population and a sample.
3. Distinguish between experimental and non-experimental research.
4. Distinguish between lab studies, field studies, and field experiments.

Identifying and Defining the Variables and Population

Variables and Operational Definitions

Part of generating a hypothesis involves identifying the variables that you want to study and operationally defining those variables so that they can be measured. Research questions in psychology are about variables. A **variable** is a quantity or quality that varies across people or situations. For example, the height of the students enrolled in a university course is a variable because it varies from student to student. The chosen major of the students is also a variable as long as not everyone in the class has declared the same major. Almost everything in our world varies and as such thinking of examples of constants (things that don't vary) is far more difficult. A rare example of a constant is the speed of light. Variables can be either quantitative or categorical. A **quantitative variable** is a quantity, such as height, that is typically measured by assigning a number to each individual. Other examples of quantitative variables include people's level of talkativeness, how depressed they are, and the number of siblings they have. A **categorical variable** is a quality, such as chosen major, and is typically measured by assigning a category label to each individual (e.g., Psychology, English, Nursing, etc.). Other examples include people's nationality, their occupation, and whether they are receiving psychotherapy.

After the researcher generates their hypothesis and selects the variables they want to manipulate and measure, the researcher needs to find ways to actually measure the variables of interest. This requires an **operational definition**—a definition of the variable in terms of precisely how it is to be measured. Most variables that researchers are interested in studying cannot be directly observed or measured and this poses a problem because empiricism (observation) is at the heart of the scientific method. Operationally defining a variable involves taking an abstract construct like depression that cannot be directly observed and transforming it into something that can be directly observed and measured. Most variables can be

operationally defined in many different ways. For example, depression can be operationally defined as people's scores on a paper-and-pencil depression scale such as the Beck Depression Inventory, the number of depressive symptoms they are experiencing, or whether they have been diagnosed with major depressive disorder. Researchers are wise to choose an operational definition that has been used extensively in the research literature.

Sampling and Measurement

In addition to identifying which variables to manipulate and measure, and operationally defining those variables, researchers need to identify the population of interest. Researchers in psychology are usually interested in drawing conclusions about some very large group of people. This is called the **population**. It could be all American teenagers, children with autism, professional athletes, or even just human beings—depending on the interests and goals of the researcher. But they usually study only a small subset or **sample** of the population. For example, a researcher might measure the talkativeness of a few hundred university students with the intention of drawing conclusions about the talkativeness of men and women in general. It is important, therefore, for researchers to use a representative sample—one that is similar to the population in important respects.

One method of obtaining a sample is **simple random sampling**, in which every member of the population has an equal chance of being selected for the sample. For example, a pollster could start with a list of all the registered voters in a city (the population), randomly select 100 of them from the list (the sample), and ask those 100 whom they intend to vote for. Unfortunately, random sampling is difficult or impossible in most psychological research because the populations are less clearly defined than the registered voters in a city. How could a researcher give all American teenagers or all children with autism an equal chance of being selected for a sample? The most common alternative to random sampling is **convenience sampling**, in which the sample consists of individuals who happen to be nearby and willing to participate (such as introductory psychology students). Of course, the obvious problem with convenience sampling is that the sample might not be representative of the population and therefore it may be less appropriate to generalize the results from the sample to that population.

Experimental vs. Non-Experimental Research

The next step a researcher must take is to decide which type of approach they will use to collect the data. As you will learn in your research methods course there are many different approaches to research that can be divided in many different ways. One of the most fundamental distinctions is between experimental and non-experimental research.

Experimental Research

Researchers who want to test hypotheses about causal relationships between variables (i.e., their goal is to explain) need to use an experimental method. This is because the experimental method is the only method that allows us to determine causal relationships. Using the experimental approach, researchers first manipulate one or more variables while attempting to control extraneous variables, and then they measure how the manipulated variables affect participants' responses.

The terms independent variable and dependent variable are used in the context of experimental research. The **independent variable** is the variable the experimenter manipulates (it is the presumed cause) and the **dependent variable** is the variable the experimenter measures (it is the presumed effect).

Extraneous variables are any variable other than the dependent variable. **Confounds** are a specific type of extraneous variable that systematically varies along with the variables under investigation and therefore provides an alternative explanation for the results. When researchers design an experiment they need to ensure that they control for confounds; they need to ensure that extraneous variables don't become confounding variables because in order to make a causal conclusion they need to make sure alternative explanations for the results have been ruled out.

As an example, if we manipulate the lighting in the room and examine the effects of that manipulation on workers' productivity, then the lighting conditions (bright lights vs. dim lights) would be considered the independent variable and the workers' productivity would be considered the dependent variable. If the bright lights are noisy then that noise would be a confound since the noise would be present whenever the lights are bright and the noise would be absent when the lights are dim. If noise is varying systematically with light then we wouldn't know if a difference in worker productivity across the two lighting conditions is due to noise or light. So confounds are bad, they disrupt our ability to make causal conclusions about the nature of the relationship between variables. However, if there is noise in the room both when the lights are on and when the lights are off then noise is merely an extraneous variable (it is a variable other than the independent or dependent variable) and we don't worry much about extraneous variables. This is because unless a variable varies systematically with the manipulated independent variable it cannot be a competing explanation for the results.

Non-Experimental Research

Researchers who are simply interested in describing characteristics of people, describing relationships between variables, and using those relationships to make predictions can use non-experimental research. Using the non-experimental approach, the researcher simply measures variables as they naturally occur, but they do not manipulate them. For instance, if I just measured the number of traffic fatalities in America last year that involved the use of a cell phone but I did not actually manipulate cell phone use then this would be categorized as non-experimental research. Alternatively, if I stood at a busy intersection and recorded drivers' genders and whether or not they were using a cell phone when they passed through the intersection to see whether men or women are more likely to use a cell phone when driving, then this would be non-

experimental research. It is important to point out that non-experimental does not mean nonscientific. Non-experimental research is scientific in nature. It can be used to fulfill two of the three goals of science (to describe and to predict). However, unlike with experimental research, we cannot make causal conclusions using this method; we cannot say that one variable causes another variable using this method.

Laboratory vs. Field Research

The next major distinction between research methods is between laboratory and field studies. A **laboratory study** is a study that is conducted in the laboratory environment. In contrast, a **field study** is a study that is conducted in the real-world, in a natural environment.

Laboratory experiments typically have high **internal validity**. Internal validity refers to the degree to which we can confidently infer a causal relationship between variables. When we conduct an experimental study in a laboratory environment we have very high internal validity because we manipulate one variable while controlling all other outside extraneous variables. When we manipulate an independent variable and observe an effect on a dependent variable and we control for everything else so that the only difference between our experimental groups or conditions is the one manipulated variable then we can be quite confident that it is the independent variable that is causing the change in the dependent variable. In contrast, because field studies are conducted in the real-world, the experimenter typically has less control over the environment and potential extraneous variables, and this decreases internal validity, making it less appropriate to arrive at causal conclusions.

But there is typically a trade-off between internal and external validity. **External validity** simply refers to the degree to which we can generalize the findings to other circumstances or settings, like the real-world environment. When internal validity is high, external validity tends to be low; and when internal validity is low, external validity tends to be high. So laboratory studies are typically low in external validity, while field studies are typically high in external validity. Since field studies are conducted in the real-world environment it is far more appropriate to generalize the findings to that real-world environment than when the research is conducted in the more artificial sterile laboratory.

Finally, there are field studies which are non-experimental in nature because nothing is manipulated. But there are also **field experiments** where an independent variable is manipulated in a natural setting and extraneous variables are controlled. Depending on their overall quality and the level of control of extraneous variables, such field experiments can have high external and high internal validity.

12. Analyzing the Data

Learning Objectives

1. Distinguish between descriptive and inferential statistics
2. Identify the different kinds of descriptive statistics researchers use to summarize their data
3. Describe the purpose of inferential statistics.
4. Distinguish between Type I and Type II errors.

Once the study is complete and the observations have been made and recorded the researchers need to analyze the data and draw their conclusions. Typically, data are analyzed using both descriptive and inferential statistics. Descriptive statistics are used to summarize the data and inferential statistics are used to generalize the results from the sample to the population. In turn, inferential statistics are used to make conclusions about whether or not a theory has been supported, refuted, or requires modification.

Descriptive Statistics

Descriptive statistics are used to organize or summarize a set of data. Examples include percentages, measures of central tendency (mean, median, mode), measures of dispersion (range, standard deviation, variance), and correlation coefficients.

Measures of central tendency are used to describe the typical, average and center of a distribution of scores. The **mode** is the most frequently occurring score in a distribution. The **median** is the midpoint of a distribution of scores. The **mean** is the average of a distribution of scores.

Measures of dispersion are also considered descriptive statistics. They are used to describe the degree of spread in a set of scores. So are all of the scores similar and clustered around the mean or is there a lot of variability in the scores? The **range** is a measure of dispersion that measures the distance between the highest and lowest scores in a distribution. The **standard deviation** is a more sophisticated measure of dispersion that measures the average distance of scores from the mean. The **variance** is just the standard deviation squared. So it also measures the distance of scores from the mean but in a different unit of measure.

Typically means and standard deviations are computed for experimental research studies in which an independent variable was manipulated to produce two or more groups and a dependent variable was

measured quantitatively. The means from each experimental group or condition are calculated separately and are compared to see if they differ.

For non-experimental research, simple percentages may be computed to describe the percentage of people who engaged in some behavior or held some belief. But more commonly non-experimental research involves computing the correlation between two variables. A **correlation coefficient** describes the strength and direction of the relationship between two variables. The values of a correlation coefficient can range from -1.00 (the strongest possible negative relationship) to +1.00 (the strongest possible positive relationship). A value of 0 means there is no relationship between the two variables. Positive correlation coefficients indicate that as the values of one variable increase, so do the values of the other variable. A good example of a positive correlation is the correlation between height and weight, because as height increases weight also tends to increase. Negative correlation coefficients indicate that as the value of one variable increase, the values of the other variable decrease. An example of a negative correlation is the correlation between stressful life events and happiness; because as stress increases, happiness is likely to decrease.

Inferential Statistics

As you learned in the section of this chapter on sampling, typically researchers sample from a population but ultimately they want to be able to generalize their results from the sample to a broader population. Researchers typically want to infer what the population is like based on the sample they studied. Inferential statistics are used for that purpose. **Inferential statistics** allow researchers to draw conclusions about a population based on data from a sample. Inferential statistics are crucial because the effects (i.e., the differences in the means or the correlation coefficient) that researchers find in a study may be due simply to random chance variability or they may be due to a real effect (i.e., they may reflect a real relationship between variables or a real effect of an independent variable on a dependent variable).

Researchers use inferential statistics to determine whether their effects are statistically significant. A **statistically significant** effect is one that is unlikely due to random chance and therefore likely represents a real effect in the population. More specifically results that have less than a 5% chance of being due to random error are typically considered statistically significant. When an effect is statistically significant it is appropriate to generalize the results from the sample to the population. In contrast, if inferential statistics reveal that there is more than a 5% chance that an effect could be due to chance error alone then the researcher must conclude that their result is not statistically significant.

It is important to keep in mind that statistics are probabilistic in nature. They allow researchers to determine whether the chances are low that their results are due to random error, but they don't provide any absolute certainty. Hopefully, when we conclude that an effect is statistically significant it is a real effect that we would find if we tested the entire population. And hopefully when we conclude that an effect is not statistically significant there really is no effect and if we tested the entire population we would find no effect. And that 5% threshold is set at 5% to ensure that there is a high probability that we make a correct decision and that our determination of statistical significance is an accurate reflection of reality.

But mistakes can always be made. Specifically, two kinds of mistakes can be made. First, researchers can

make a **Type I error**, which is a false positive. It is when a researcher concludes that their results are statistically significant (so they say there is an effect in the population) when in reality there is no real effect in the population and the results are just due to chance (they are a fluke). When the threshold is set to 5%, which is the convention, then the researcher has a 5% chance or less of making a Type I error. You might wonder why researchers don't set it even lower to reduce the chances of making a Type I error. The reason is when the chances of making a Type I error are reduced, the chances of making a Type II error are increased. A **Type II error** is a missed opportunity. It is when a researcher concludes that their results are not statistically significant when in reality there is a real effect in the population and they just missed detecting it. Once again, these Type II errors are more likely to occur when the threshold is set too low (e.g., set at 1% instead of 5%) and/or when the sample was too small.

13. Drawing Conclusions and Reporting the Results

Learning Objectives

1. Identify the conclusions researchers can make based on the outcome of their studies.
2. Describe why scientists avoid the term “scientific proof.”
3. Explain the different ways that scientists share their findings.

Drawing Conclusions

Since statistics are probabilistic in nature and findings can reflect type I or type II errors, we cannot use the results of a single study to conclude with certainty that a theory is true. Rather theories are supported, refuted, or modified based on the results of research.

If the results are statistically significant and consistent with the hypothesis and the theory that was used to generate the hypothesis, then researchers can conclude that the theory is supported. Not only did the theory make an accurate prediction, but there is now a new phenomenon that the theory accounts for. If a hypothesis is disconfirmed in a systematic empirical study, then the theory has been weakened. It made an inaccurate prediction, and there is now a new phenomenon that it does not account for.

Although this seems straightforward, there are some complications. First, confirming a hypothesis can strengthen a theory but it can never prove a theory. In fact, scientists tend to avoid the word “prove” when talking and writing about theories. One reason for this avoidance is that the result may reflect a type I error. Another reason for this avoidance is that there may be other plausible theories that imply the same hypothesis, which means that confirming the hypothesis strengthens all those theories equally. A third reason is that it is always possible that another test of the hypothesis or a test of a new hypothesis derived from the theory will be disconfirmed. This difficulty is a version of the famous philosophical “problem of induction.” One cannot definitively prove a general principle (e.g., “All swans are white.”) just by observing confirming cases (e.g., white swans)—no matter how many. It is always possible that a disconfirming case (e.g., a black swan) will eventually come along. For these reasons, scientists tend to think of theories—even highly successful ones—as subject to revision based on new and unexpected observations.

A second complication has to do with what it means when a hypothesis is disconfirmed. According to the strictest version of the hypothetico-deductive method, disconfirming a hypothesis disproves the theory it was derived from. In formal logic, the premises “if A then B” and “not B” necessarily lead to the conclusion

“not A.” If A is the theory and B is the hypothesis (“if A then B”), then disconfirming the hypothesis (“not B”) must mean that the theory is incorrect (“not A”). In practice, however, scientists do not give up on their theories so easily. One reason is that one disconfirmed hypothesis could be a missed opportunity (the result of a type II error) or it could be the result of a faulty research design. Perhaps the researcher did not successfully manipulate the independent variable or measure the dependent variable.

A disconfirmed hypothesis could also mean that some unstated but relatively minor assumption of the theory was not met. For example, if Zajonc had failed to find social facilitation in cockroaches, he could have concluded that drive theory is still correct but it applies only to animals with sufficiently complex nervous systems. That is, the evidence from a study can be used to modify a theory. This practice does not mean that researchers are free to ignore disconfirmations of their theories. If they cannot improve their research designs or modify their theories to account for repeated disconfirmations, then they eventually must abandon their theories and replace them with ones that are more successful.

The bottom line here is that because statistics are probabilistic in nature and because all research studies have flaws there is no such thing as scientific proof, there is only scientific evidence.

Reporting the Results

The final step in the research process involves reporting the results. As described in the section on Reviewing the Research Literature in this chapter, results are typically reported in peer-reviewed journal articles and at conferences.

The most prestigious way to report one's findings is by writing a manuscript and having it published in a peer-reviewed scientific journal. Manuscripts published in psychology journals typically must adhere to the writing style of the American Psychological Association (APA style). You will likely be learning the major elements of this writing style in this course.

Another way to report findings is by writing a book chapter that is published in an edited book. Preferably the editor of the book puts the chapter through peer review but this is not always the case and some scientists are invited by editors to write book chapters.

A fun way to disseminate findings is to give a presentation at a conference. This can either be done as an oral presentation or a poster presentation. Oral presentations involve getting up in front of an audience of fellow scientists and giving a talk that might last anywhere from 10 minutes to 1 hour (depending on the conference) and then fielding questions from the audience. Alternatively, poster presentations involve summarizing the study on a large poster that provides a brief overview of the purpose, methods, results, and discussion. The presenter stands by their poster for an hour or two and discusses it with people who pass by. Presenting one's work at a conference is a great way to get feedback from one's peers before attempting to undergo the more rigorous peer-review process involved in publishing a journal article.

14. Key Takeaways and Exercise

Key Takeaways

- Research in psychology can be described by a simple cyclical model. A research question based on the research literature leads to an empirical study, the results of which are published and become part of the research literature.
- The research literature in psychology is all the published research in psychology, consisting primarily of articles in professional journals and scholarly books.
- Early in the research process, it is important to conduct a review of the research literature on your topic to refine your research question, identify appropriate research methods, place your question in the context of other research, and prepare to write an effective research report.
- There are several strategies for finding previous research on your topic. Among the best is using PsycINFO, a computer database that catalogs millions of articles, books, and book chapters in psychology and related fields.
- Research questions expressed in terms of variables and relationships between variables can be suggested by other researchers or generated by asking a series of more general questions about the behavior or psychological characteristic of interest.
- It is important to evaluate how interesting a research question is before designing a study and collecting data to answer it. Factors that affect interestingness are the extent to which the answer is in doubt, whether it fills a gap in the research literature, and whether it has important practical implications.
- It is also important to evaluate how feasible a research question will be to answer. Factors that affect feasibility include time, money, technical knowledge and skill, and access to special equipment and research participants.
- A theory is broad in nature and explains larger bodies of data. A hypothesis is more specific and makes a prediction about the outcome of a particular study.
- Working with theories is not “icing on the cake.” It is a basic ingredient of psychological research.
- Like other scientists, psychologists use the hypothetico-deductive method. They construct theories to explain or interpret phenomena (or work with existing theories), derive hypotheses from their theories, test the hypotheses, and then reevaluate the theories in light of the new results.
- Variables vary across people or situations and may be quantitative (e.g., age) or categorical (e.g., course subject).
- A sample is a small subset of a larger population that is selected to participate in the research study. There are many different ways of sampling participants including convenience sampling and simple random sampling.
- Experimental research involves manipulating an independent variable to observe the effects on a measured dependent variable while non-experimental research involves measuring variables as they naturally occur (i.e., without manipulating anything).
- Research can be conducted in the field or the lab. Laboratory experiments tend to have high internal validity (allowing us to make strong causal conclusions), while field studies often have more external validity (allowing us to generalize to the real world).
- The mean, median, and mode are measures of central tendency used to describe the typical, average, or

center scores in a distribution. The range, standard deviation, and variance are measures of how dispersed or spread apart the scores are. Measures of central tendency and dispersion are important descriptive statistics.

- Inferential statistics allow researchers to determine whether their findings are statistically significant, that is, whether they are unlikely to be due to chance alone and therefore are likely to represent a real effect in the population.
- Since statistics are probabilistic in nature we never know if our conclusions are correct. We can make type I errors (concluding an effect is real when it is not) or type II errors (concluding there is no effect when there actually is a real effect in the population).
- Theories can be supported by not proved. Similarly, disconfirming a hypothesis does not necessarily mean that theory has been disproved.
- The final step of the research process involves reporting results at scientific conferences, in journal articles, and/or in books.

Exercises

- Practice: Find a description of an empirical study in a professional journal or in one of the scientific psychology blogs. Then write a brief description of the research in terms of the cyclical model presented here. One or two sentences for each part of the cycle should suffice.
- Watch the following TED Ed video, in which David H. Schwartz provides an introduction to two types of empirical studies along with some methods that scientists use to increase the reliability of their results:



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://kpu.pressbooks.pub/psychmethods4e/?p=271#oembed-1>

- Reading in print? Go to <https://youtu.be/GUPd2HJHUt8> or scan this QR code with your phone:



- Practice: Use the techniques discussed in this section to find 10 journal articles and book chapters on one of the following research ideas: memory for smells, aggressive driving, the causes of narcissistic personality disorder, the functions of the intraparietal sulcus, or prejudice against the physically handicapped.
- Watch the following video clip produced by UBCiSchool about how to read an academic paper (without

losing your mind):



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://kpu.pressbooks.pub/psychmethods4e/?p=271#oembed-2>

- Reading in print? Go to https://youtu.be/SKxm2HF_-k0 or scan this QR code with your phone:



- Practice: Generate three research ideas based on each of the following: informal observations, practical problems, and topics discussed in recent issues of professional journals.
- Practice: Generate an empirical research question about each of the following behaviors or psychological characteristics: long-distance running, getting tattooed, social anxiety, bullying, and memory for early childhood events.
- Practice: Evaluate each of the research questions you generated in Exercise 2 in terms of its interestingness based on the criteria discussed in this section.
- Practice: Find an issue of a journal that publishes short empirical research reports (e.g., *Psychological Science*, *Psychonomic Bulletin and Review*, *Personality and Social Psychology Bulletin*). Pick three studies, and rate each one in terms of how feasible it would be for you to replicate it with the resources available to you right now. Use the following rating scale: (1) You could replicate it essentially as reported. (2) You could replicate it with some simplifications. (3) You could not replicate it. Explain each rating.
- Practice: Find a recent empirical research report in a professional journal. Read the introduction and highlight in different colors descriptions of theories and hypotheses.
- Practice: Using the research article you found in a professional journal identify whether the study was experimental or non-experimental. If it was experimental identify the independent and dependent variables.
- Practice: Using the research article you found in a professional journal identify which descriptive statistics were reported.
- Practice: Describe why theories can be supported but not proved.

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